



CHALLENGES IN FEDERATING EDGE RESOURCES

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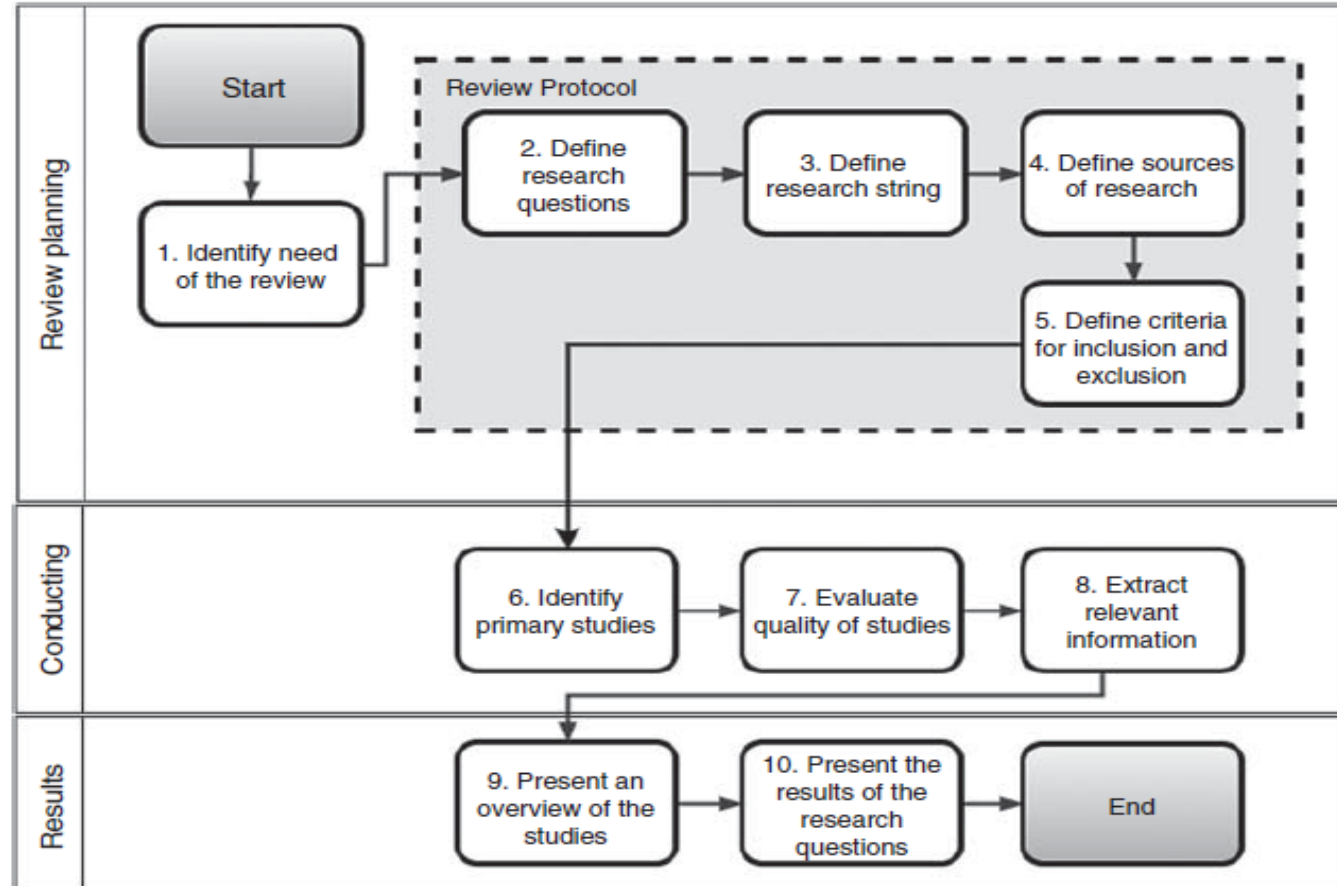
INTRODUCTION

- Cisco estimates that there are IoE, with estimates of 8–10 billion connected today.
- Cloud computing has been a key enabling technology for the IoT, a small increase in the percentage of connected or cyber-physical objects represents dramatic change in the feature space of computing and a potential tsunami of computation and hyper-connectivity, which today's infrastructure will struggle to accommodate at historic levels of quality of service (QoS).
- Traditional cloud computing architectures were simply not designed with an IoT, characterized by extreme geographic distribution, heterogeneity and dynamism, in mind.
- A novel approach is required to meet the requirements of IoT including transversal requirements (scalability, interoperability, flexibility, reliability, efficiency, availability, and security) as well as cloud-to-thing (C2T)-specific computation, storage and communication needs

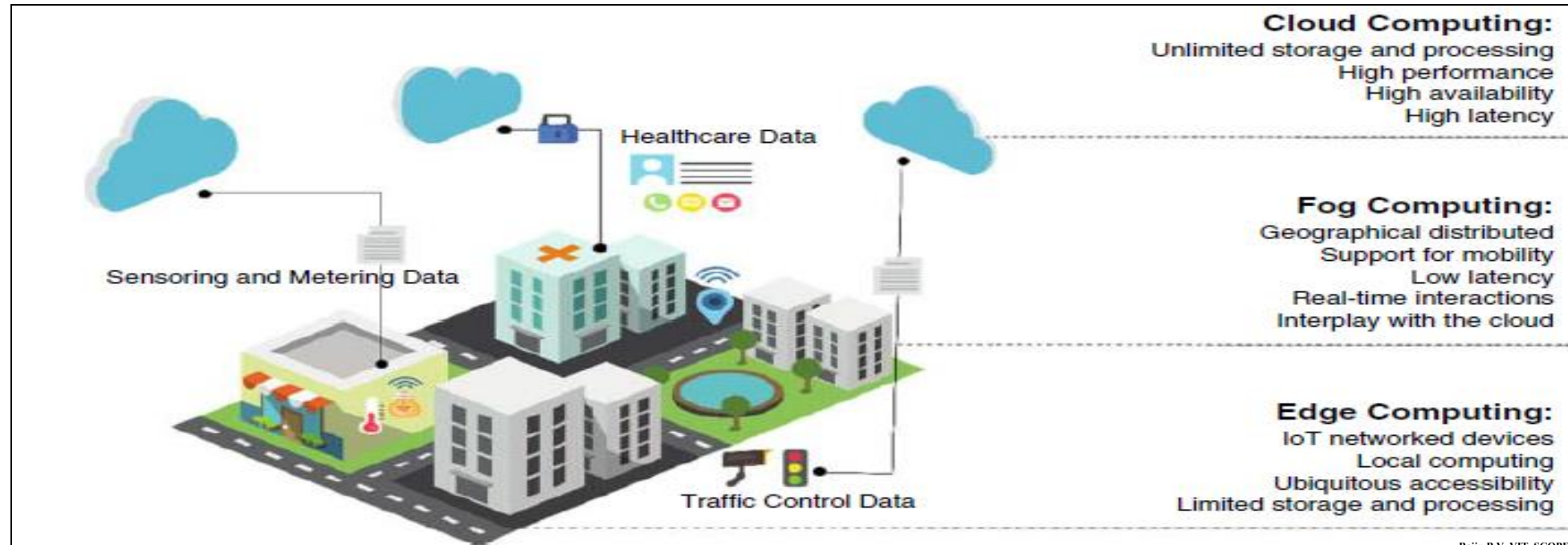
METHODOLOGY

- The objective of this systematic literature review is to present an overview of academic literature on

- (i) Use of modeling techniques to represent and evaluate an integrated C2F2T system*
- (ii) Main scenarios modeled*
- (iii) Metrics used to evaluate models.*

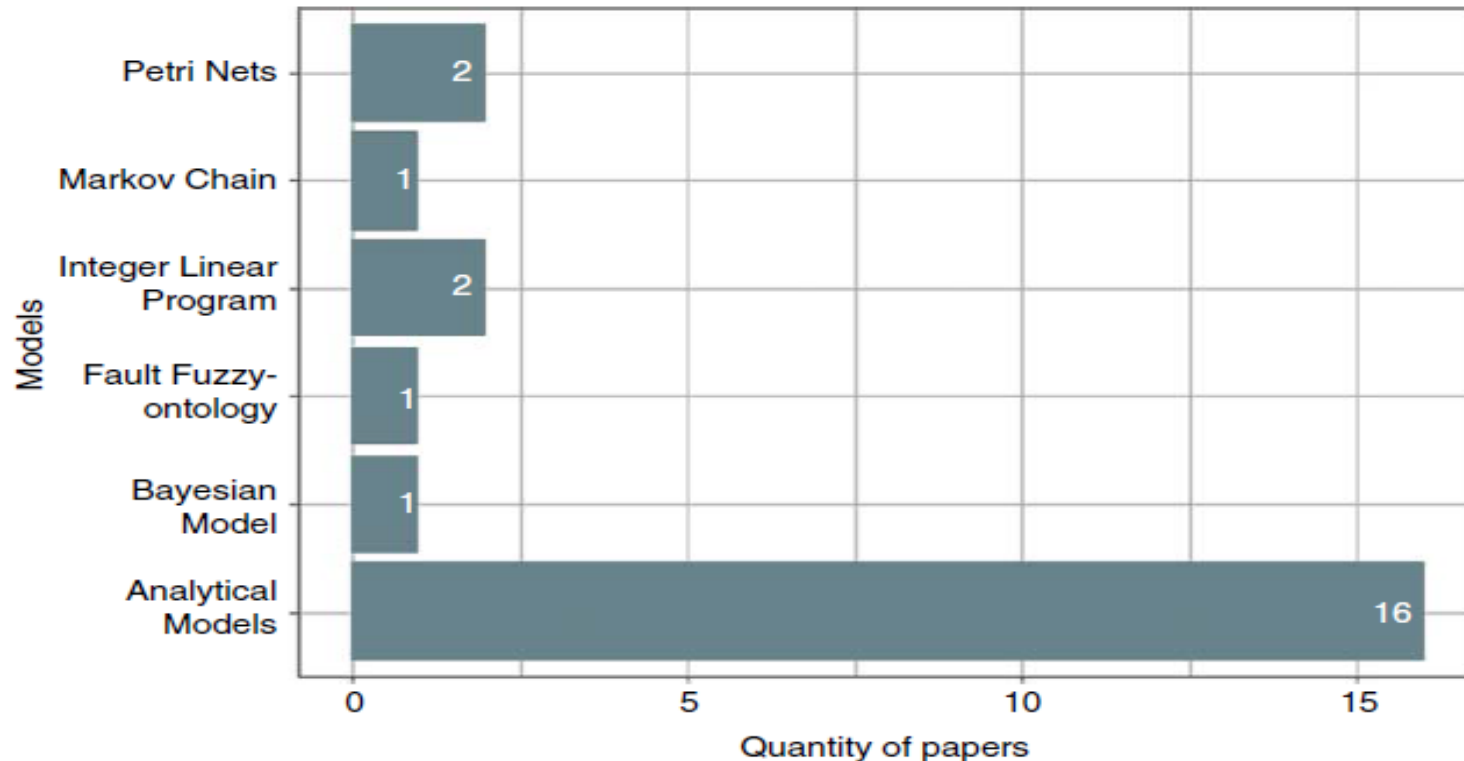


- **Fog computing** is a technology that acts as a middle layer between the cloud and smart devices (like sensors and smartphones).
- It uses **fog nodes** (intermediate devices) to manage data and communication, making it easier to run Internet of Things (IoT) applications.
- In essence, fog computing helps process data closer to where it's generated, reducing the need to send everything to the cloud.
- This makes IoT applications faster and more efficient.



Integrated C2F2T Literature by Modeling Technique

- The analysis presented suggests that **analytical models** followed by **Petri Nets** and **Integer Linear Programs** are the most common techniques used for modeling an integrated C2F2T system.



1.1 Analytical Models

- Analytical models are mathematical models that have a **closed-form solution**, i.e. the solution to the equations used to describe changes in a given system can be expressed as a mathematical analytic function.
- Analytical models can be used to predict computing resource requirements related to

- *workload behavior*
- *content*
- *volume changes*

and to measure effects of hardware and software changes.

- In this systematic review 16 articles are using analytical models.

(1) Wei Li et al. (System modelling and performance evaluation of a three-tier cloud of things) defined the architecture, where each physical and virtual component of layers are described as a vector of associated features.

- A set of equations is defined to calculate metrics such as **power consumed** and **time latency** of scenarios.

(2) K. E. Desikan et al. (A Novel Distributed Latency-Aware Data Processing in Fog Computing-Enabled IoT Networks) , considered the

- **number of gateways** (devices that receive data from IoT devices and forward to cloud infrastructure or fog devices) present in an architecture.
- **total data received** for each gateway.
- **time spent** by each gateway to process the data, in their modeling.

Imagine we have a fog computing network with 3 gateways (G1, G2, G3) that receive data from IoT devices and process it before sending it to the cloud.

Gateway 1 (G1)

Total Data Received: 100 MB

Processing Time: 10 seconds

Gateway 2 (G2)

Total Data Received: 150 MB

Processing Time: 15 seconds

Gateway 3 (G3)

Total Data Received: 200 MB

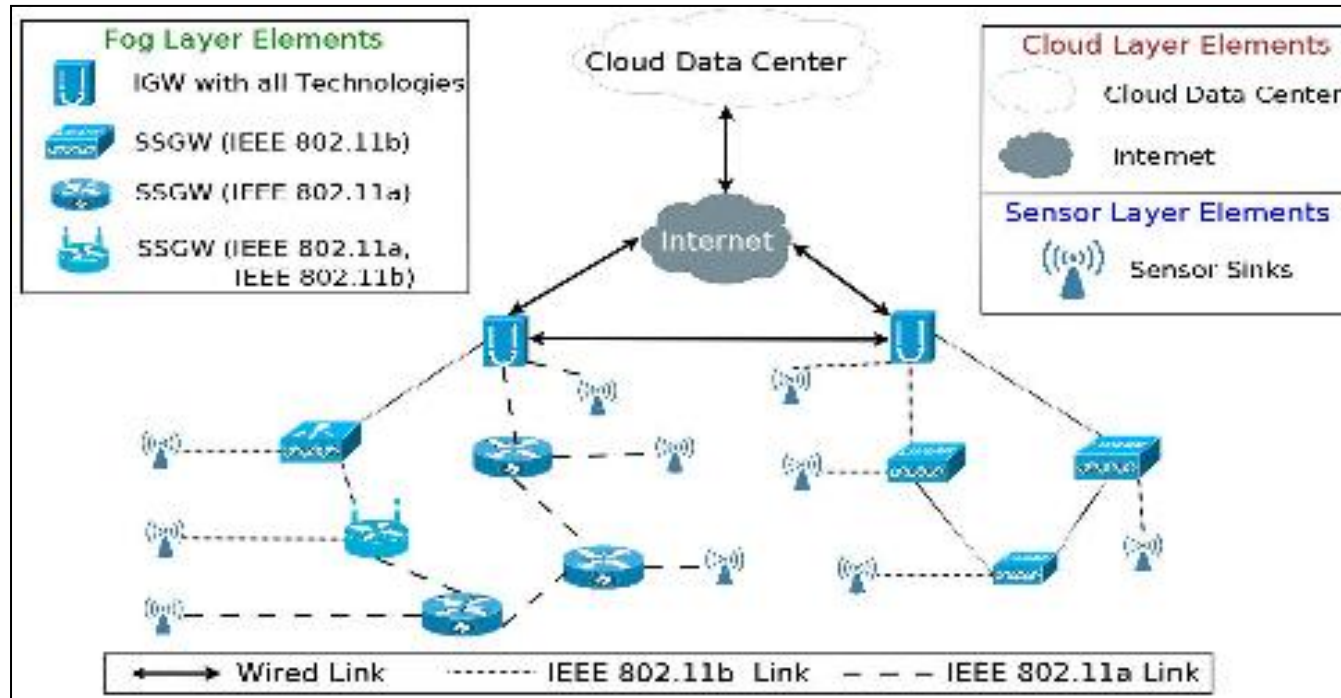
Processing Time: 20 seconds

G1 processes 100 MB of data in 10 seconds.

G2 processes 150 MB of data in 15 seconds.

G3 processes 200 MB of data in 20 seconds.

- Equations are proposed to represent the **available buffer** and the **occupancy efficiency** of gateway buffers.
- Depending on the available space on the gateways, the data are transferred to higher layers, increasing the delay for data processing.
- Several equations are defined for the calculation of the delay and the authors propose an optimization of the gateways' efficiency, improving the occupancy and the response time efficiency for all gateways in the system.



(3) J.F. Colom et al. (Collaborative building of behavioural models based on Internet of Things) proposed an analytical model to represent a **scheduling mechanism** in IoT environment.

- An equation is defined to represent the addition of a load to be processed in a device.
- This process takes into account several variables, such as **processor load**, **free memory**, and **free bandwidth**, among others.

(4) H.Wu et al. (Deviation-based neighborhood model for context-aware QoS prediction of cloud and IoT Services) proposed a method to **predict** the **QoS** of IoT services.

- The model is based on a neighborhood collaborative filter and allows an efficient global optimization scheme.
- A set of equations is defined to predict correctly the QoS of a service, based on aspects such as **latency**, **response time**, and **user network condition**.

Thermostat

Processor Load (PL): 20%

Free Memory (FM): 50MB

Free Bandwidth (FB): 10Mbps

Security Camera

Processor Load (PL): 50%

Free Memory (FM): 100MB

Free Bandwidth (FB): 20Mbps

Smart Light

Processor Load (PL): 10%

Free Memory (FM): 30MB

Free Bandwidth (FB): 5Mbps

Let's say we want to add a new load that requires:

Processor Load: 15%

Memory: 20MB

Bandwidth: 5Mbps

Thermostat

New Processor Load: (20% + 15% = 35%)

New Free Memory: (50MB - 20MB = 30MB)

New Free Bandwidth: (10Mbps - 5Mbps = 5Mbps)

Security Camera

New Processor Load: (50% + 15% = 65%)

New Free Memory: (100MB - 20MB = 80MB)

New Free Bandwidth: (20Mbps - 5Mbps = 15Mbps)

Smart Light

New Processor Load: (10% + 15% = 25%)

New Free Memory: (30MB - 20MB = 10MB)

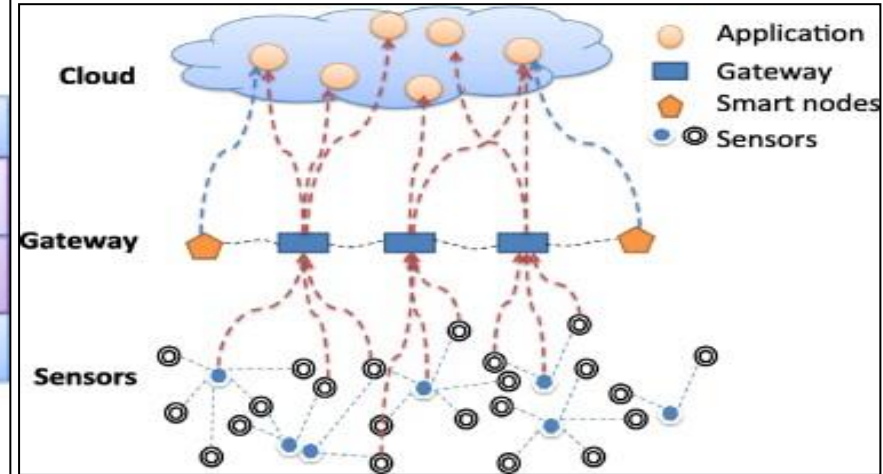
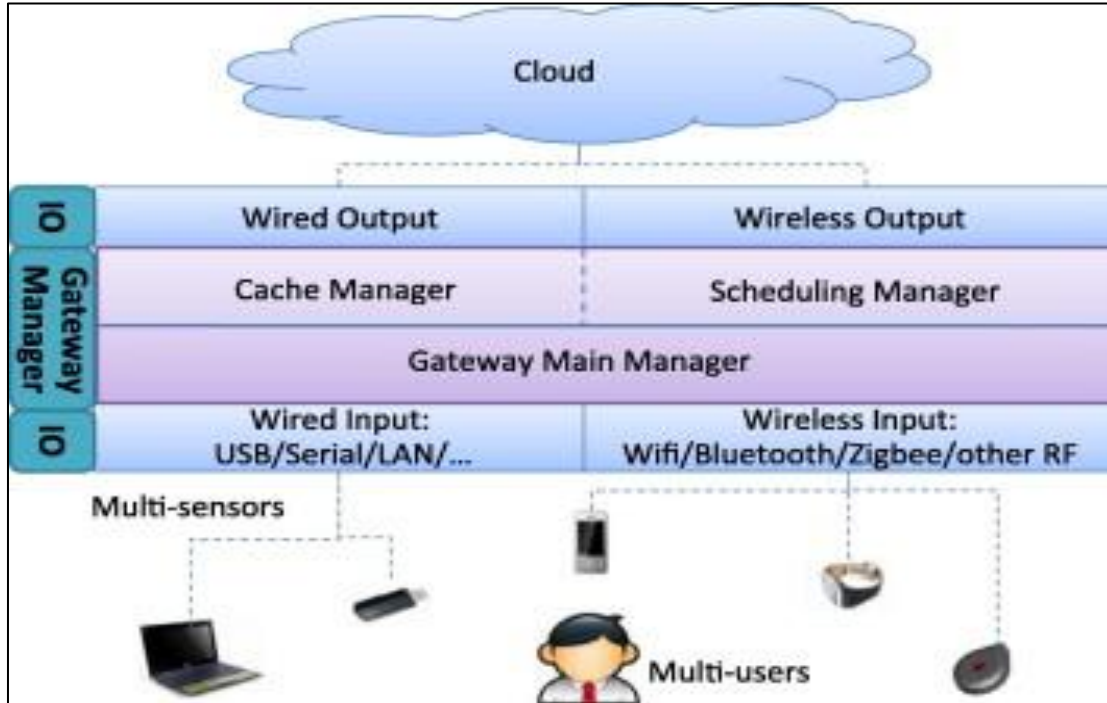
New Free Bandwidth: (5Mbps - 5Mbps = 0Mbps)

Thermostat: New load is manageable.

Security Camera: New load is manageable.

Smart Light: New load might be too high due to limited bandwidth.

(5) **Y. Lyu, F. Yan *et.al*** (High-performance scheduling model for multisensor gateway of cloud sensor system-based smart-living) proposed a **scheduling model** to manage sensor applications by a gateway.



- The requirement resources of applications are taken into account in scheduling problem.

- In a scenario with n applications

$$A = \{a_i(R_i, r_i, w_i(t), s_i(t))\}$$

where R_i is a **resource requirement vector**

$r_i \in [0, 1]$ is a **priority**

$w_i(t)$ is a **required work status**

$s_i(t)$ is an **actual working status** of a_i

- The integral calculates the total waiting time of the tasks, where
 - T is **evaluation time**.
 - u_c is **total CPU utilization**
 - c_m is **total memory capacity**
 - b_l **input bandwidth**
 - b_o **output bandwidth**
- Vector $[\alpha_c, \alpha_m, -\alpha_l, \alpha_o]$ represents the overloading factors for the **CPU**, **memory**, **input** and **output bandwidth**, respectively, and indicates that the system can run smoothly to some extent.

$$\min.\varphi = \sum_{i=1}^n \int_0^T r_i w_i(t) [1 - s_i(t)] dt$$

$$s. t. (1) \sum_{i=1}^n u_i s_i(t) \leq \alpha_c u_c$$

$$(2) \sum_{i=1}^n c_i s_i(t) \leq \alpha_m c_m$$

$$(3) \sum_{i=1}^n b_i^l s_i(t) \leq \alpha_l b_l$$

$$(4) \sum_{i=1}^n b_i^o s_i(t) \leq \alpha_o b_o$$

(6) F. Renna et al. (Query processing for the Internet-of-Things: Coupling of device energy consumption and cloud infrastructure billing), proposed a set of analytic expressions for representing the **expected energy consumption of devices**, as well as **a cloud billing method** for a group of devices on an IoT aggregator.

- Considering n devices the below equation computes the **energy consumption of each device** over the **monitoring period**, T .

$$E_{\text{exp}} = E[\Psi_e]g_e + i_e \int_0^{c_e E[\Psi_e]} (c_e E[\Psi_e] - \omega_e) P_e(\omega_e) d\omega_e$$

where $E[\Psi_e]$ is a **query data volume of devices**,

g_e is **consumption rate** (joule-per-bit)

i_e is **“idle” energy consumption by each device**.

- The integral of the second term represents the **expected energy consumption of a device** in **idle mode**.
 - $c_e E[\Psi_e]$ is a threshold where the application activates “idle” mode
 - $P_e(\omega_e)$ represent the **probability density function** of Ψ_e .

$E[\Psi_e]$: Let's assume $E[\Psi_e]=2$

g_e, i_e, c_e : Let $g_e=3$, $i_e=2$ and $c_e=4$

$$E_{\text{exp}} = E[\Psi_e]g_e + i_e \int_0^{c_e E[\Psi_e]} (c_e E[\Psi_e] - \omega_e) P_e(\omega_e) d\omega_e$$

$P_e(\omega_e)$: Assume $P_e(\omega_e)$ is uniform between $[0, c_e E[\Psi_e]]$

$$P_e(\omega_e) = \frac{1}{c_e E[\Psi_e]} = \frac{1}{8}$$

Compute the first term: $E[\Psi_e]g_e = 2 \cdot 3 = 6$

Compute the second term: $\int_0^{c_e E[\Psi_e]} (c_e E[\Psi_e] - \omega_e) P_e(\omega_e) d\omega_e.$

Substituting $P_e(\omega_e)=1/8$ and $c_e E[\Psi_e]=8$

$$\int_0^8 (8 - \omega_e) \frac{1}{8} d\omega_e.$$

Integral part split into 2

$$\frac{1}{8} \int_0^8 (8 - \omega_e) d\omega_e = \frac{1}{8} \left[\int_0^8 8 d\omega_e - \int_0^8 \omega_e d\omega_e \right]$$

$$\frac{1}{8} (64 - 32) = \frac{1}{8} \cdot 32 = 4.$$

The integral of a constant a over the range $[0, b]$ is: $\int_0^b a dx = a \cdot b.$ $\int_0^8 8 d\omega_e = 8 \cdot 8 = 64,$

The integral of ω_e is: $\int_0^8 \omega_e d\omega_e = \frac{\omega_e^2}{2} \Big|_0^8$ $\int_0^8 \omega_e d\omega_e = \frac{\omega_e^2}{2} \Big|_0^8 = \frac{8^2}{2} - 0 = 32.$

Multiply by $i_e=2$: $2 \cdot 4 = 8$

Total E_{exp} :

$$E_{\text{exp}} = 6 + 8 = 14$$

- The **Expected cloud billing cost** can be calculated by , taking into account n aggregated query volumes from n devices.
 - $E[\Psi_b]g_b$ represents the **transfer/storage costs**.
 - the first integral represents the **idle billing cost**
 - the second integral corresponds to the **active billing cost**

$$B_{\text{exp}} = E[\Psi_b]g_b + i_b \int_0^{c_b} (c_b - \omega_b)P(\omega_b)d\omega_b + p_b \int_{c_b}^{\infty} (\omega_b - c_b)P(\omega_b)d\omega_b$$

i_b is the **cost of transferring one bit** (dollars-per-query-bit),

c_b is a **coupling point** that define the expect billing cost.

(7) F. Jalali et al. (Interconnecting fog computing and microgrids for greening IoT) the utilization of a strategy using fog computing with microgrids to reduce the energy consumption of IoT applications is considered.

- Two equations are proposed to evaluate energy consumption.

- Energy consumption between **IoT and cloud infrastructure** is calculated using

$$E_{IoT-cloud} = E_{GW-r} + E_{GW-t} + E_{net} + E_{DC}$$

- (E_{GW-r}) energy consumed by IoT gateways when receiving data from IoT devices and sensors
- (E_{GW-t}) energy consumed by IoT gateways to transmit data to the cloud data center
- (E_{net}) energy consumed by transport network between IoT gateways and cloud
- (E_{DC}) the energy consumed by components of data center (E_{DC}).

Given Data:

Energy consumed by the Gateway Receiver: $E_{GW-r} = 5$ units

Energy consumed by the Gateway Transmitter: $E_{GW-t} = 3$ units

Energy consumed in the Network: $E_{net} = 2$ units

Energy consumed by the Data Center: $E_{DC} = 10$ units

$$E_{IoT-cloud} = 5 + 3 + 2 + 10 = 20 \text{ units}$$

- Energy consumption between IoT and fog infrastructure is calculated using

$$E_{IoT-fog} = E_{GW-r} + E_{GW-c} + \beta(E_{GW-t} + E_{net} + E_{DC})$$

- This equation takes into account the same components of previous equation plus two other components:
 - E_{GW-c} , the energy consumed by IoT gateways for local computation and processing
 - β , a ratio of the number of updates from the fog to the cloud for synchronization.

Let's assume:

$$E_{GW-r} = 4 \text{ units}$$

$$E_{GW-c} = 6 \text{ units}$$

$$E_{GW-t} = 3 \text{ units}$$

$$E_{net} = 2 \text{ units}$$

$$E_{DC} = 8 \text{ units}$$

$$\beta = 0.5 (50\% \text{ of the data is offloaded to the Cloud})$$

$$E_{IoT-fog} = 4 + 6 + 0.5(3 + 2 + 8)$$

$$E_{IoT-fog} = 16.5 \text{ units}$$

(8) R. K. Behera et al. (Reliability modelling of service-oriented Internet of Things) proposed an analytical modeling for estimating the **reliability of an IoT scenario** for a smart home context.

- An algorithm is proposed to estimate the reliability of the IoT service, which is formed by n subsystems.
- Reliability of IoT system is calculated using the below equation

$$R_s(t_b) = \prod_{i=1}^N ISR_i \times \prod_{i=1}^f P_f(i) \times \prod_{i=1}^k P_{pr}(i)$$

(P_{pr}) the availability of the all k programs running on the virtual machines

(P_f) the availability of the f input files for the programs

(ISR) the reliability of each subsystem , i.e. the reliability of the VM being executed.

(9) W. Zeng et al. (Opacity in Internet of Things with Cloud Computing) , the authors presented a model to evaluate the **security level** of C2T systems.

- The focus of the model is the flow of information, where an initial state of the system is defined, and a set of operations are performed.
- Thus, after performing these actions, the system reaches new states.

Let's assume:

$N=3$ (There are 3 components).

$ISR_1=0.9$, $ISR_2=0.85$, $ISR_3=0.95$ (Reliabilities of individual components).

$Pf(1)=0.1$, $Pf(2)=0.15$, $Pf(3)=0.05$ (Failure probabilities of the components).

$Ppr(1)=0.8$, $Ppr(2)=0.9$, $Ppr(3)=0.85$ (Performance probabilities of the components).

$$R_s(t_b) = \prod_{i=1}^3 ISR_i \times \prod_{i=1}^3 Pf(i) \times \prod_{i=1}^3 Ppr(i)$$

$$R_s(t_b) = \prod_{i=1}^N ISR_i \times \prod_{i=1}^f P_f(i) \times \prod_{i=1}^k P_{pr}(i)$$

$$\prod_{i=1}^3 ISR_i = 0.9 \times 0.85 \times 0.95 = 0.72375$$

$$\prod_{i=1}^3 Pf(i) = 0.1 \times 0.15 \times 0.05 = 0.00075$$

$$\prod_{i=1}^3 Ppr(i) = 0.8 \times 0.9 \times 0.85 = 0.612$$

The overall reliability or performance $R_s(t_b)$ is approximately:

$$R_s(t_b) \approx 0.00033$$

$$R_s(t_b) = 0.72375 \times 0.00075 \times 0.612 = 0.00033290625$$

(10) V. B. Souza et al. (Towards Distributed Service Allocation in Fog-to-Cloud (F2C) Scenarios) used a Multidimensional Knapsack Problem (MKP) to find the optimal service allocation in C2F2T scenarios.

- For this, they consider a number of application aspects:
 - **load balancing**
 - **delay**
 - **energy consumption.**
- The service allocation is defined as an MKP problem, where the objective is threefold:
 - ***minimizing the energy consumption by devices***
 - ***minimizing the overload in terms of processing capacity***
 - ***minimize the overall allocation of services in infrastructure.***

(11) Y. Liu et al. (Incentive mechanism for computation offloading using edge computing: a Stackelberg game approach) address the interactions among cloud operator and IoT service provide as optimization problem.

- They formulated analytically the problem and maximize the utilities of cloud service with the purpose of obtaining optimal payment and computation offloading.

(12) Z. Ming et al. (QoS-aware computational method for IoT composite service) proposed an analytical modeling to represent QoS of IoT composite services, taking into account such metrics as *availability*, *reliability*, and *response time*.

- An optimization algorithm is proposed to find optimal cost with QoS constraints.

(13) Y. Li et al. (Leveraging Renewable Energy in Edge Clouds for Data Stream Analysis in IoT) proposed an analytical models that is used to *decide if offloading computing will be processed in IoT devices or in cloud*, taking into account the desired QoS and energy level available in IoT devices.

(14) S. Sarkar et al. (Theoretical modelling of fog computing: a green computing paradigm to support IoT applications) proposed an analytical models to *compare the performance between the fog architecture against traditional cloud computing*.

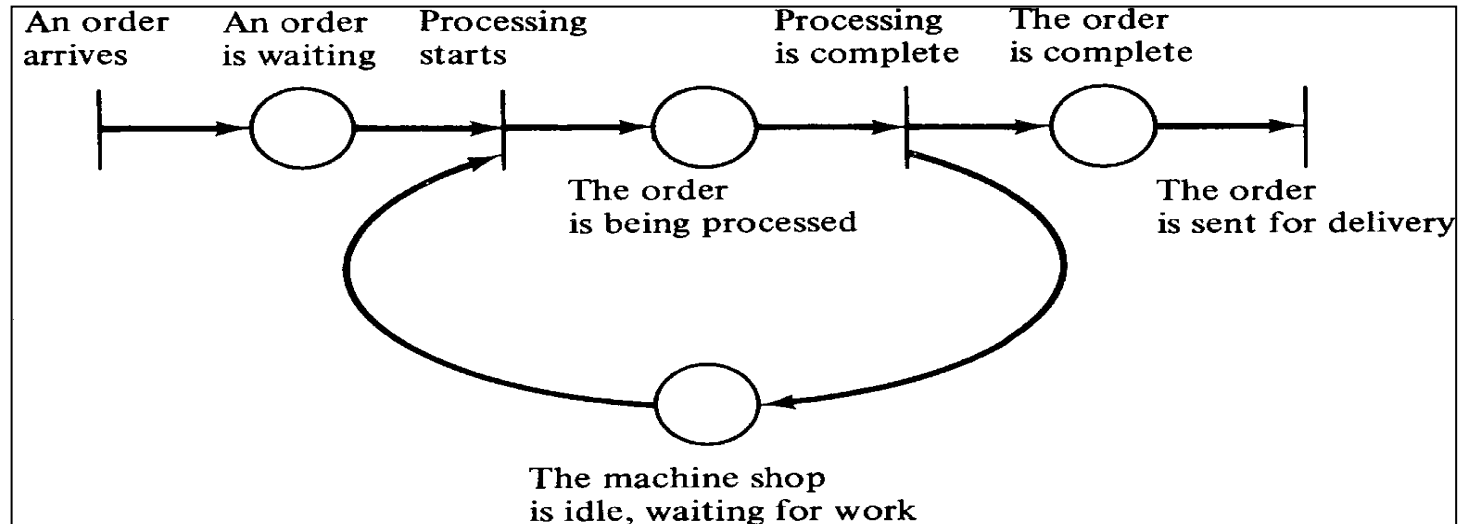
- They consider several aspects, such as
 - *location of devices*
 - *operation mode*
 - *hardware details*
 - *type of events*.

(15,16) L. Skorin-Kapov et al. (Energy efficient and quality-driven continuous sensor management for mobile IoT applications, Collaborative Computing: Networking, Applications and Worksharing (CollaborateCom)) and **T. Dinh et al.** (A location-based interactive model for Internet of Things and cloud (IoT-cloud). Ubiquitous and Future Networks) presented analytical models to represent mobile nodes connected to cloud computing. In both articles, the proposed models consider the movement of the devices that are connected to the cloud.

- In first, the model details the architecture in a selection of components: *wireless sensor network (WSN)*, the *cloud infrastructure*, *applications*, and *mobile users*.
- In second, authors consider that the *mobile devices are connected to the cells* to send data to the cloud.

1.2 Petri Net Models

- A Petri Net is a well-known model to represent systems with respect to **evaluate performance and dependability**.
- To solve a Petri Net, one can use two options:
 - (i) **analytic solution** by using Markov chain (in which case all transitions must follow exponential distributions); or
 - (ii) **simulations** using theory of discrete event simulation.
- Petri Nets allows a more fine-grained representation of the system, utilizing Markovian and non-Markovian distributions, and represents the system behavior with a fewer number of states.



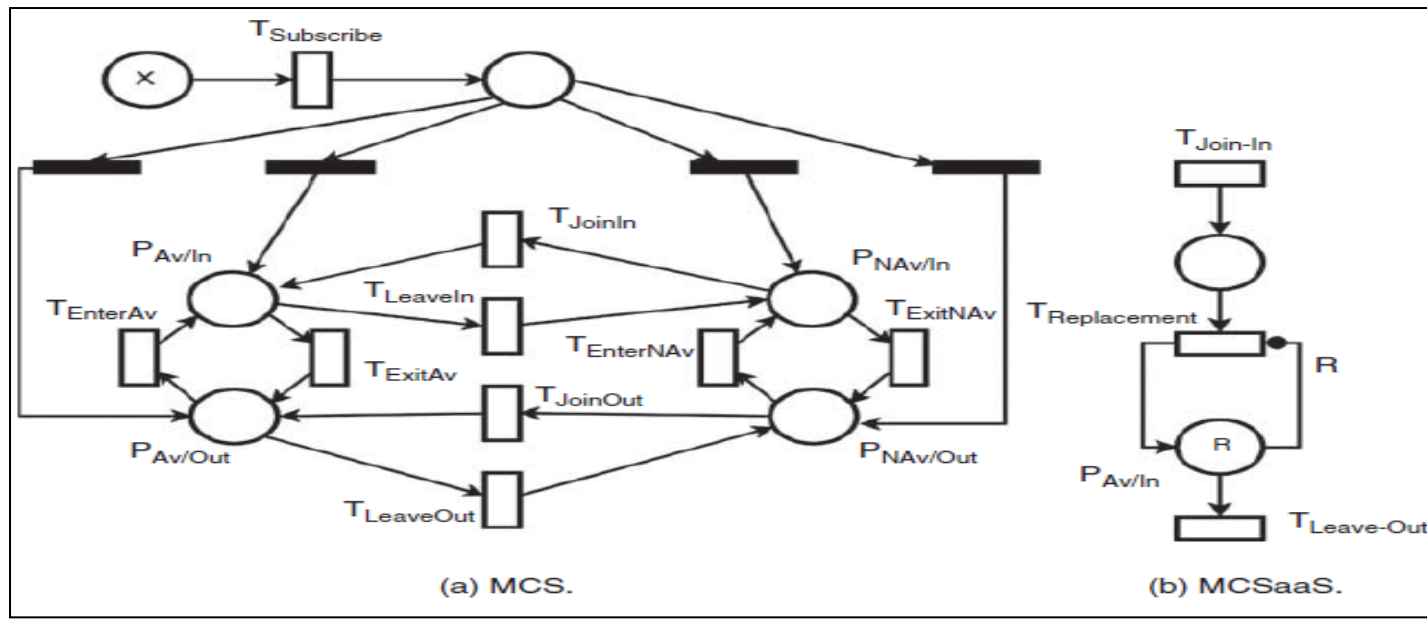
- Petri Net models used two works
- **G. Merlino** proposed a *mobile crowd sensing framework*, integrating mobile devices into services hosted in the cloud.
- This model enables efficient data collection and processing while optimizing resource utilization.
- The places represent four possible states of a contributing node:
 - *contributing node availability (Av/ NAv)*
 - *position on the interest area (In/Out).*

Available (Av): The node is available to contribute data.

Not Available (NAv): The node is not available to contribute data.

In: The node is within the area of interest.

Out: The node is outside the area of interest



The **transitions** represent the probability of the devices entering or exiting these states.

State	Description
Av/In	The node is available and located within the area of interest, ready to contribute data.
Av/Out	The node is available but outside the area of interest, thus not contributing.
NAv/In	The node is in the area of interest but unavailable due to constraints (e.g., low battery).
NAv/Out	The node is both unavailable and outside the area of interest.

- **R. K. Lomotey (Wearable IoT data stream traceability in a distributed health information system)** evaluated the problem of managing the traceability of data in C2T scenario.
- A Petri Net model is proposed **to map and match device data to users** that assists tracking a transparent data trace route, and possible detection of data compromises.
- The PetriNet represents the behavior of a proposed wearable IoT architecture.
- The **places** represent the different sources where data are generated or collected.
- **Transitions** represent events that may occur, such as new medical data readings, for example vitals, from the wearable IoT device.

Imagine a health monitoring system where wearable devices track patients' vital signs, such as heart rate and blood pressure.

Places:

Wearable Device: The source where data is generated (e.g., a smartwatch).

Health Database: Where the collected data is stored.

User Profile: The patient's profile that receives and displays the data.

Transitions:

New Reading: An event where the wearable device records a new heart rate or blood pressure reading.

Data Upload: The event of transferring the new reading from the wearable device to the health database.

Profile Update: The event of updating the patient's profile with the new data.

1.3 Integer Linear Programming

- In this systematic review, two works modeled their problems using ILP and MILP
- **E. Sturzinger** (**Application-aware resource provisioning in a heterogeneous Internet of Things**) proposed an ILP model for calculating the **financial cost of a fog-cloud architecture** located in a metropolitan area network (**MAN**).
- The authors represent application profiles and characteristics of each node in the MAN as two vectors.
- The ILP model minimizes the operational cost necessary to support the traffic in network topology while satisfy the application constraints.
- **Objective function used in ILP model** presented using equation

$$Cost_t = Cost_p + Cost_s + Cost_u + Cost_d + Cost_c$$

$Cost_t$ represents the **total cost**

$Cost_p$ is the **processing cost**

$Cost_s$ is the **storage cost**

$Cost_u$ are the **total upstream and downstream costs** respectively

$Cost_c$ is the **total MAN link capacity**.

Imagine a smart city network with three nodes: Node A, Node B, and Node C. Each node has different roles and associated costs.

Cost_p (Processing Cost): The cost of processing data at the nodes.

Node A processes traffic data: \$50

Node B processes air quality data: \$30

Node C processes energy usage data: \$20

Cost_s (Storage Cost): The cost of storing data at the nodes.

Node A stores traffic data: \$10

Node B stores air quality data: \$15

Node C stores energy usage data: \$5

Cost_u (Upload Cost): The cost of uploading data to the cloud.

Node A uploads traffic data: \$25

Node B uploads air quality data: \$20

Node C uploads energy usage data: \$15

Cost_d (Download Cost): The cost of downloading data from the cloud.

Node A downloads traffic data: \$5

Node B downloads air quality data: \$10

Node C downloads energy usage data: \$5

Cost_c (Communication Cost): The cost of communication between nodes.

Node A communicates with Node B: \$10

Node B communicates with Node C: \$10

Node C communicates with Node A: \$10

$$Cost_t = Cost_p + Cost_s + Cost_u + Cost_d + Cost_c$$

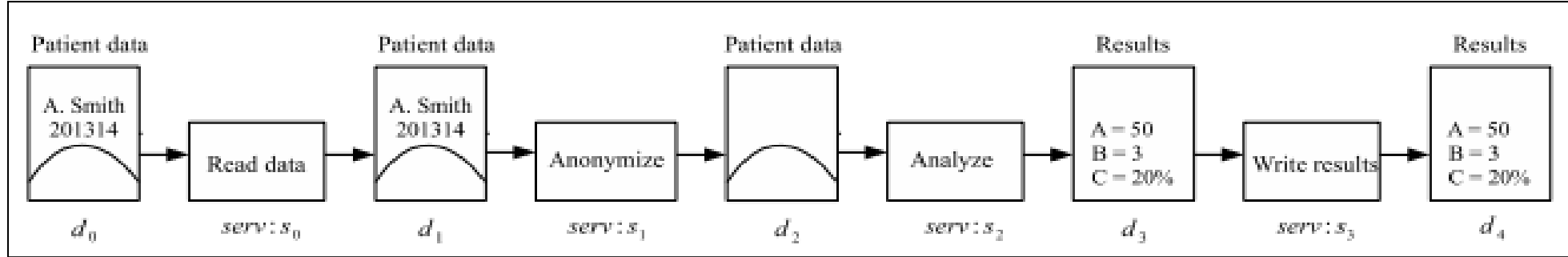
$$Cost_t = (50 + 30 + 20) + (10 + 15 + 5) + (25 + 20 + 15) + (5 + 10 + 5) + (10 + 10 + 10) = 240$$

Integrated C2F2T Literature by Use-Case Scenarios

- The use case scenarios modeled are described in the identified papers e.g. resource management, smart cities, WSN, health and other/generic are summarized in table

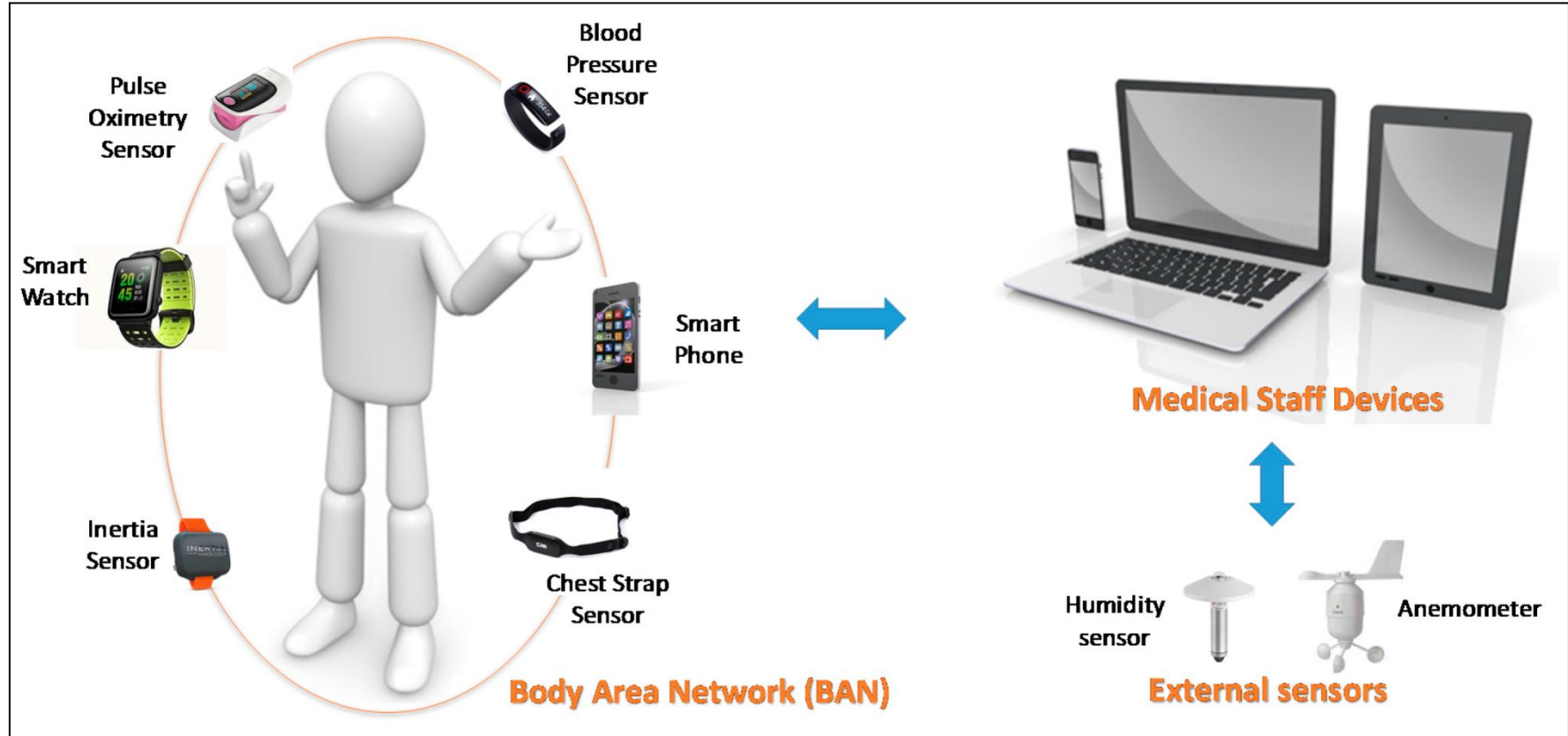
Scenarios	Applications
Generic	Generic IoT/fog/cloud applications Edge offloading computing
Health	Health monitoring Medical research
WSN	Sensing as a service Fault tolerance
Smart cities	Mobile crowd sensing as a service Smart traffic Smart living Airplane monitoring
Resource management	Resource allocation Energy efficiency Quality of service (QoS)

- **W. Zeng** (Opacity in Internet of Things with Cloud Computing) used a **medical application** as a case study for their proposed model that analyzes the **security of the information flow** in IoT systems integrated with cloud infrastructures.

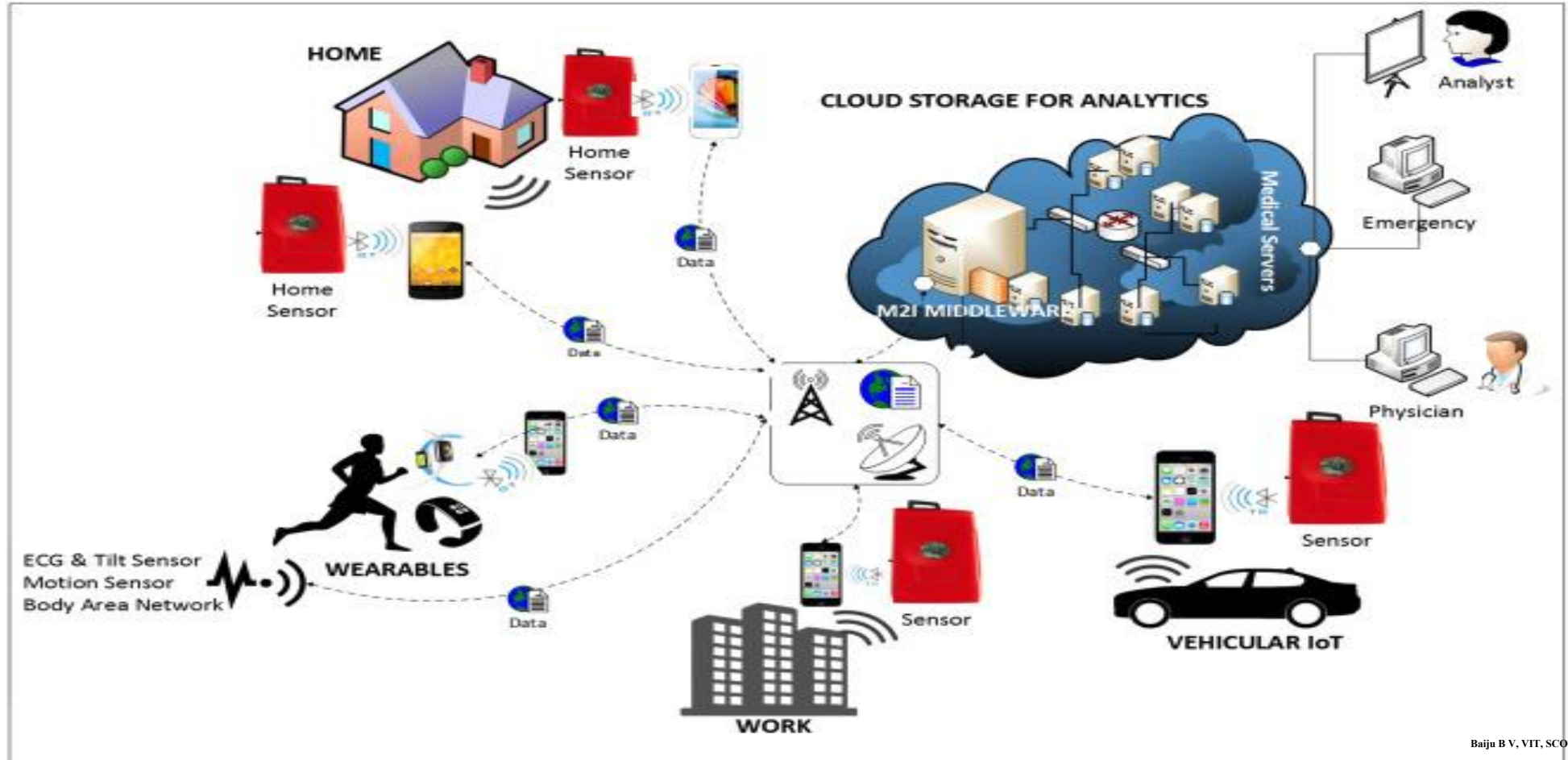


In a smart hospital, wearable IoT devices like fitness bands and smartwatches collect health data such as heart rate, oxygen levels, and blood pressure from patients. This data is sent to a shared edge server, where it is processed for different uses. For example, one application can monitor the vitals and send emergency alerts if an issue like an irregular heartbeat is detected. Another application can track long-term health trends, while a third can analyze sleep patterns or stress levels. By sharing the same IoT system, these applications work together efficiently, reducing costs and providing real-time, scalable health monitoring.

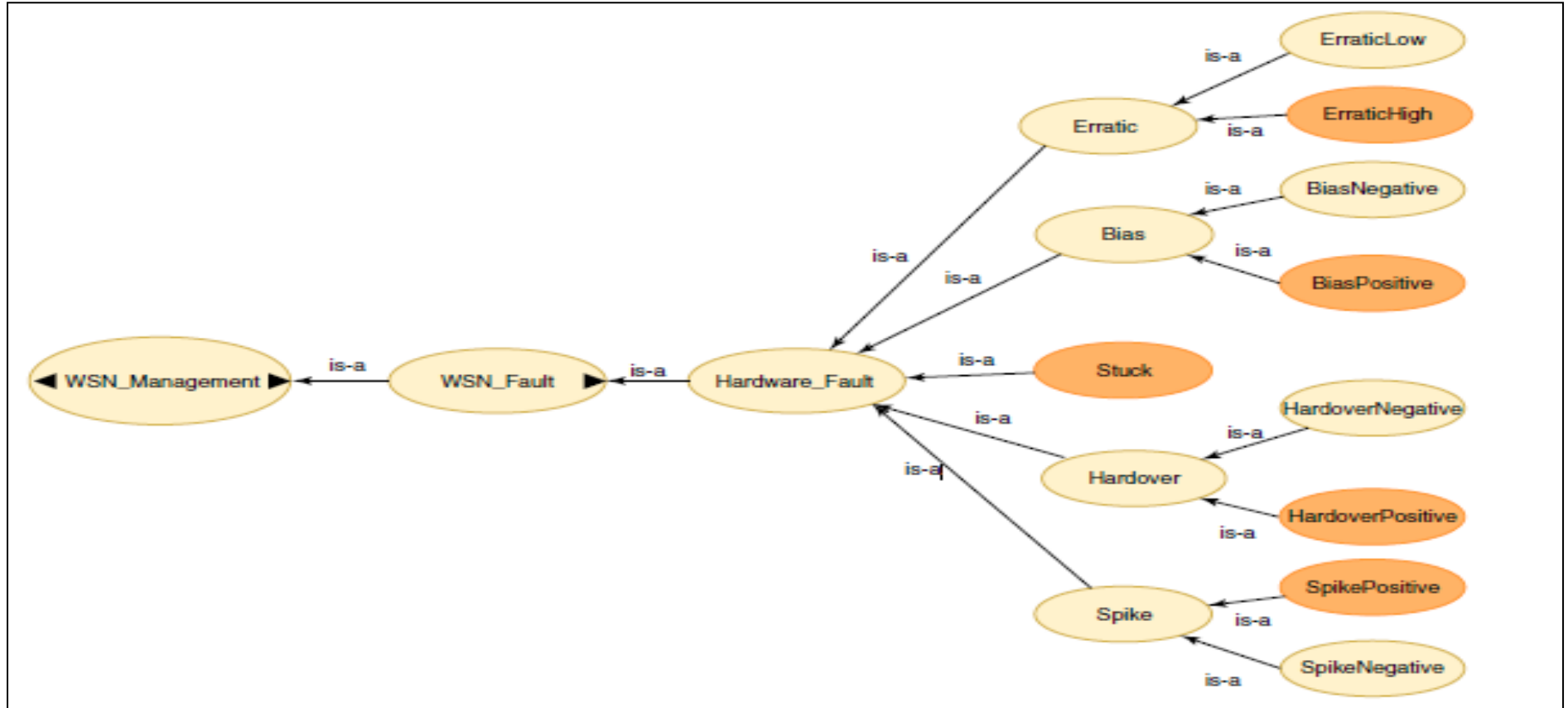
- **J.F. Colom** (Collaborative building of behavioural models based on **internet of things**) proposed a framework that **enables multiple applications** to share IoT computational devices for **health monitoring**.



- **R. K. Lomotey** (Wearable IoT data stream traceability in a distributed health information system) proposed a **wearable IoT architecture** for healthcare systems.

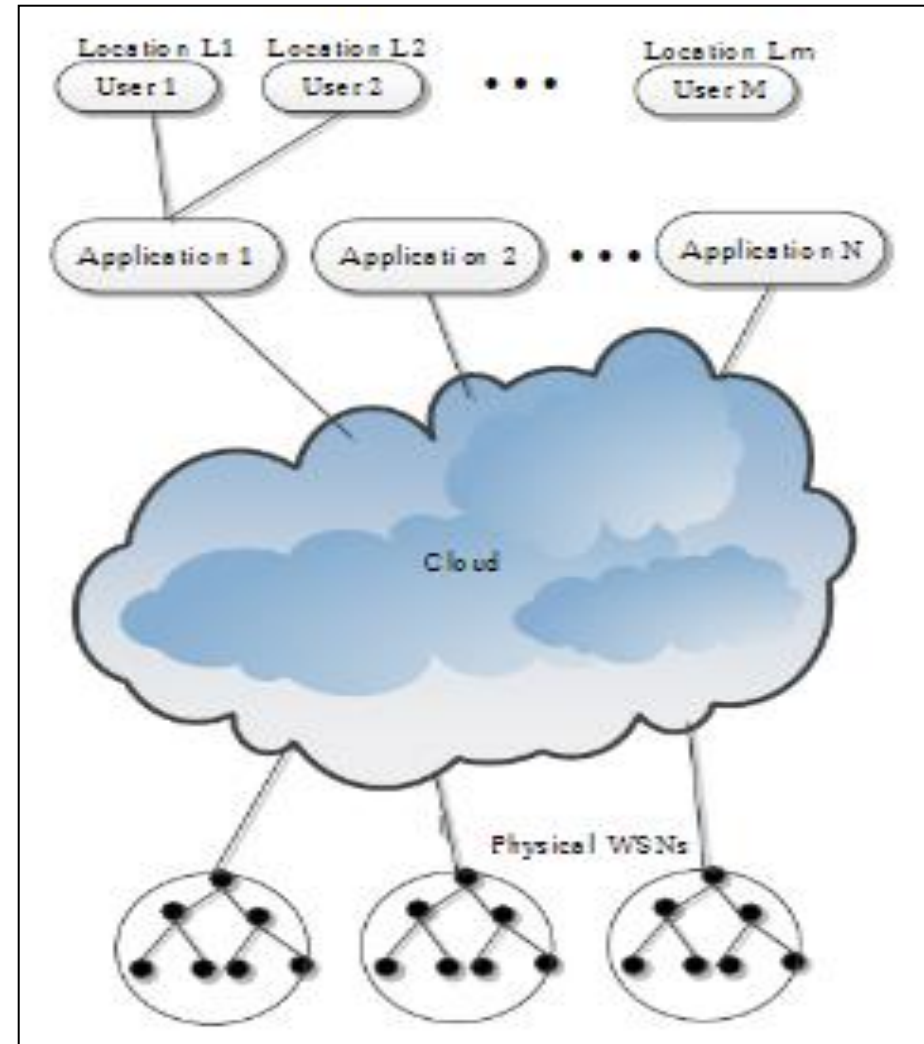


- **Y Benazzouz** (A fault fuzzy-ontology for large scale fault-tolerant wireless sensor networks) proposed a fault fuzzy-ontology that can be used to **verify fault tolerance** in large-scale WSN using service-oriented applications

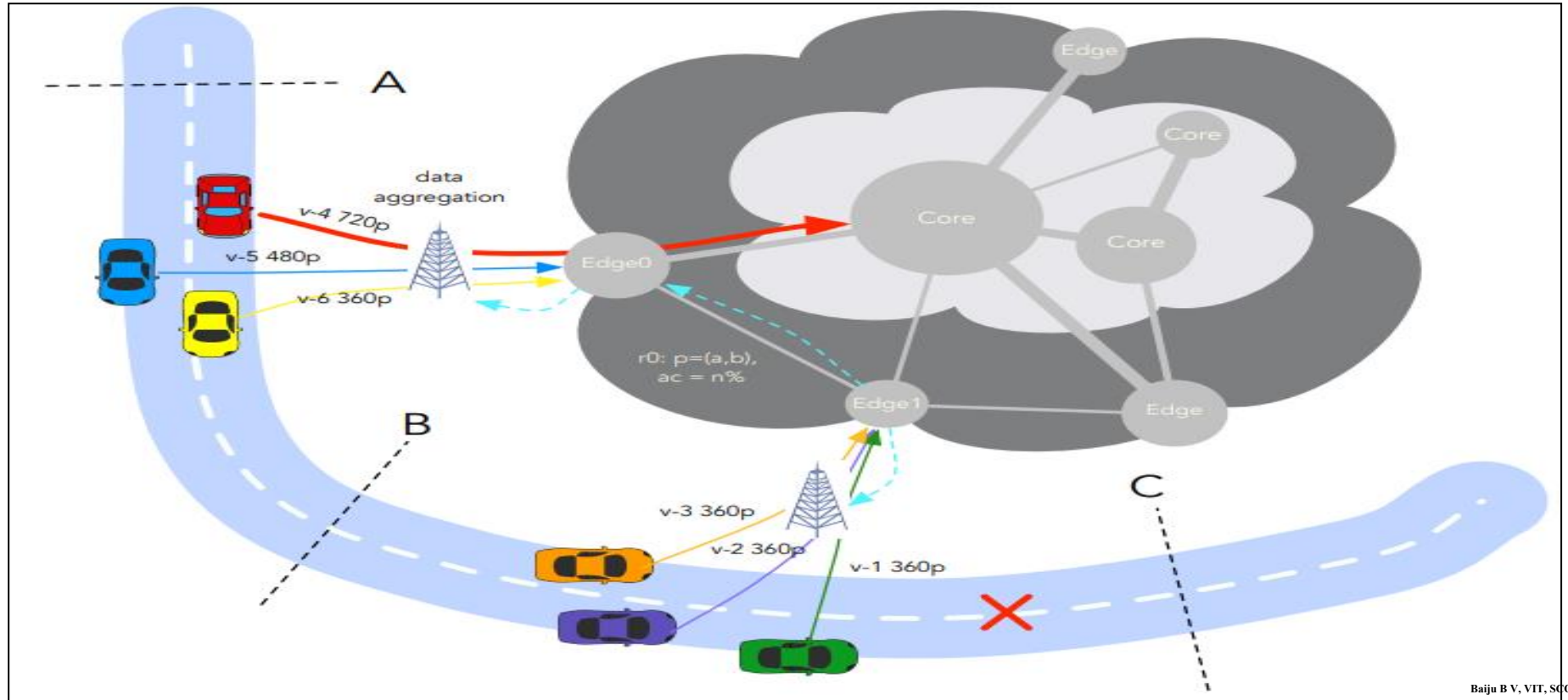


- **T. Dinh** (A location-based interactive model for Internet of Things and cloud (IoT-cloud)) proposed a WSN model for **sensing as a service** that integrates IoT and cloud infrastructures.

In a traditional model, these sensors would continuously collect and transmit data to a central server, regardless of whether anyone needs the data at that moment. This leads to unnecessary energy consumption and data transmission.

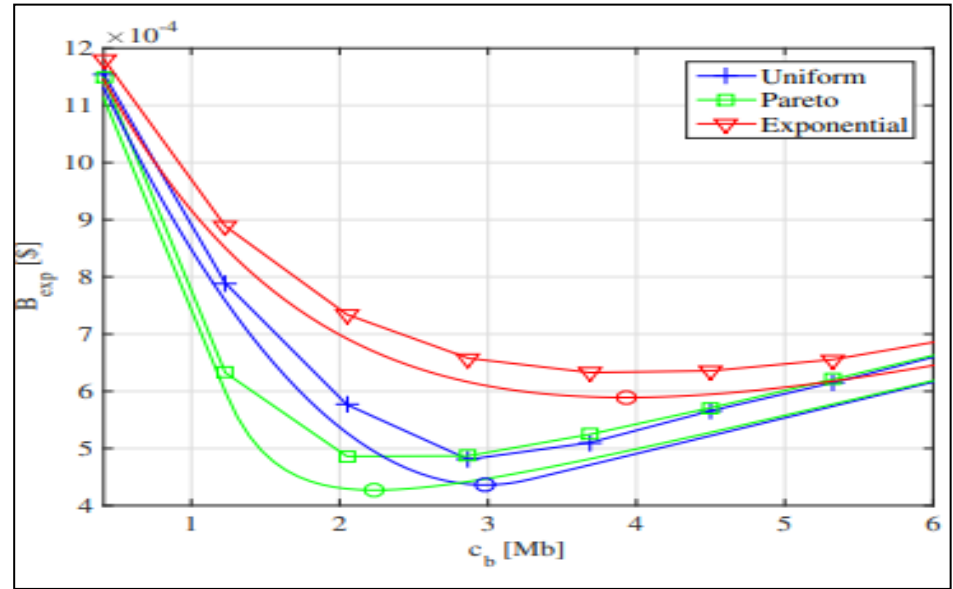
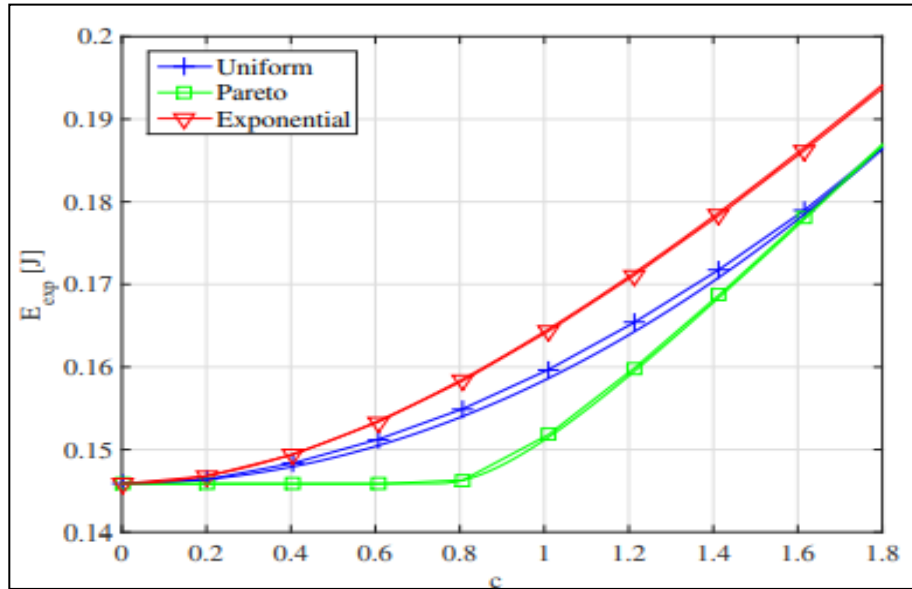


- **Y. Li . (Leveraging Renewable Energy in Edge Clouds for Data Stream Analysis in IoT)** presented a **vehicle-to-cloud monitoring application** that interacts with edge computing; their analytic model defines where the data will be processed – e.g. in the edge or in the cloud infrastructure.
- They use a **video-streaming application** to validate their model



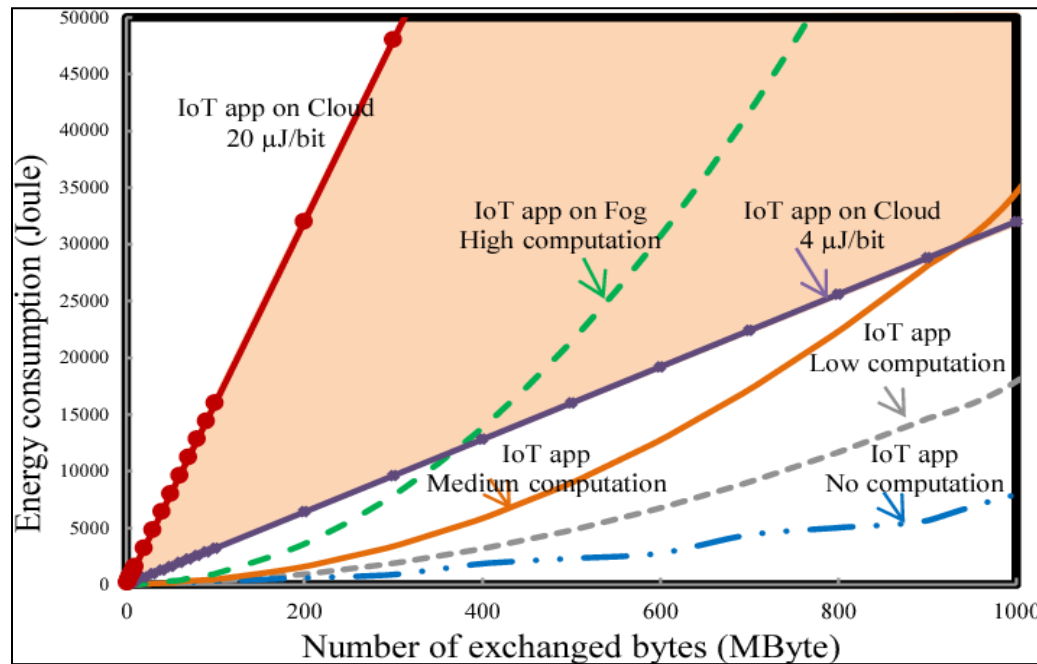
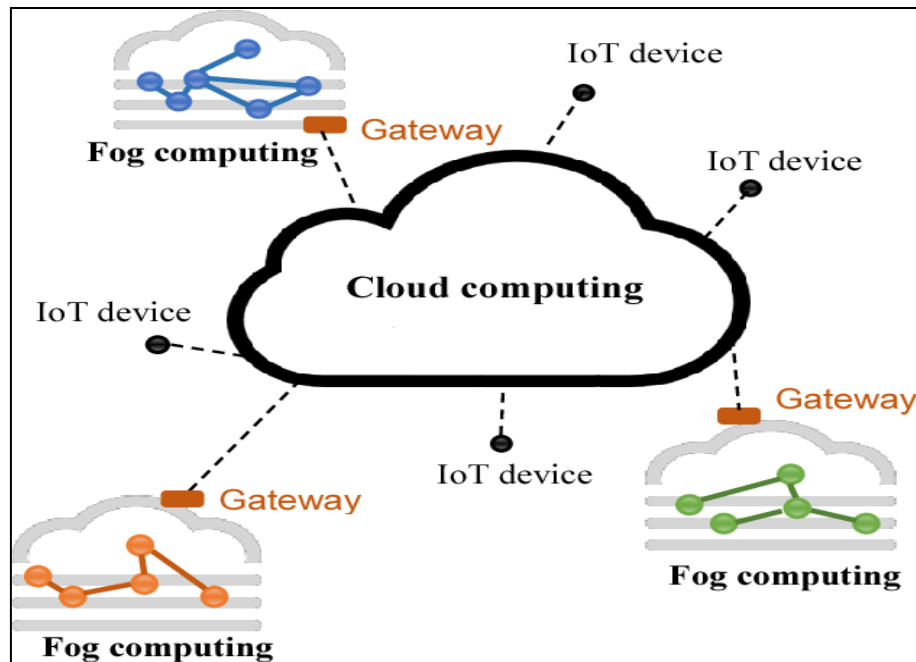
- **B. Karakostas** (Event prediction in an IoT Environment using Naïve Bayesian models) proposed an **airplane monitoring scenario**, where data generated by C2T application are analyzed through a model in order to estimate **flights delay**.
 - Suppose an inbound flight is delayed due to bad weather.
 - The model collects this data and uses the Naïve Bayesian algorithm to estimate the probability of a delay in the corresponding outbound flight.
- **Y. Lyu** (High-performance scheduling model for multisensor gateway of cloud sensor system-based smart-living) used a **smart living space** in their **scheduling model** with gateways to access cloud resources.
- **Benazzouz and I. Parissis** (A fault fuzzy-ontology for large scale fault-tolerant wireless sensor networks) presents a model to **improve the energy efficiency** of the IoT devices.
 - Suppose a sensor monitoring air quality fails.
 - The fuzzy-ontology model detects this fault by analyzing the data patterns and comparing them with expected behavior.

- **Renna et al.** (Query processing for the Internet-of-Things: Coupling of device energy consumption and cloud infrastructure billing) analyzes the **relation between energy consumption** and **cloud infrastructure billing cost**.



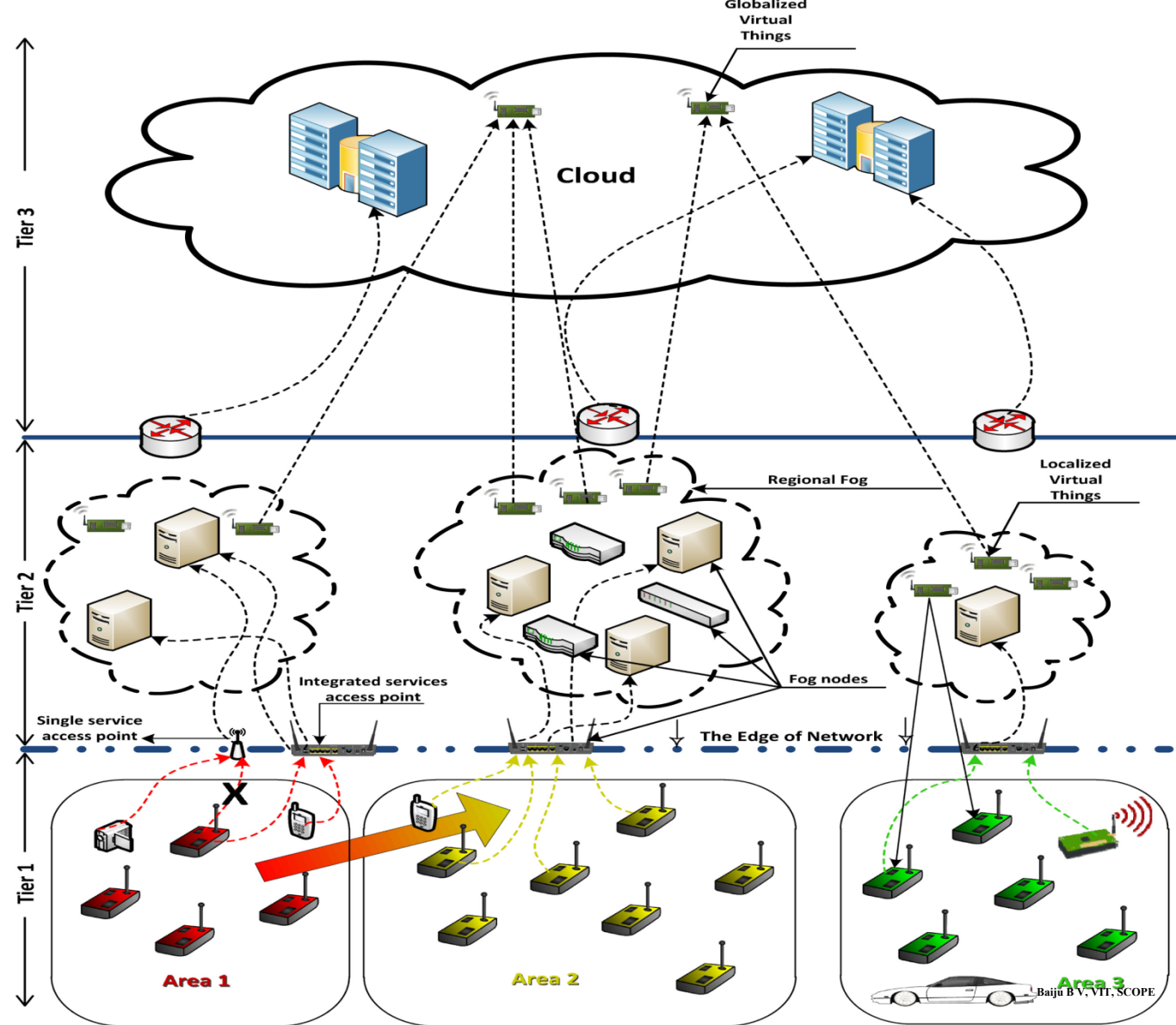
- **H.Wu et al.** (Deviation-based neighborhood model for context-aware QoS prediction of cloud and IoT Services) used a model to **analyze the energy consumption** of a WSN.

- **F. Jalali** (Interconnecting fog computing and microgrids for greening IoT) studies *energy consumption* but, in this case, the authors analyze how the integration of IoT and fog can reduce this consumption.



- **Desikan, Srinivasan, and Murthy** (A Novel Distributed Latency-Aware Data Processing in Fog Computing-Enabled IoT Networks) proposed a model to *improve the response time of* an IoT system that uses gateways to communicate to the cloud, using fog computing as an intermediary platform.

- **Li et al.** (System modelling and performance evaluation of a three-tier cloud of things) presents a **three-tier model** that address the C2F2T scenario.



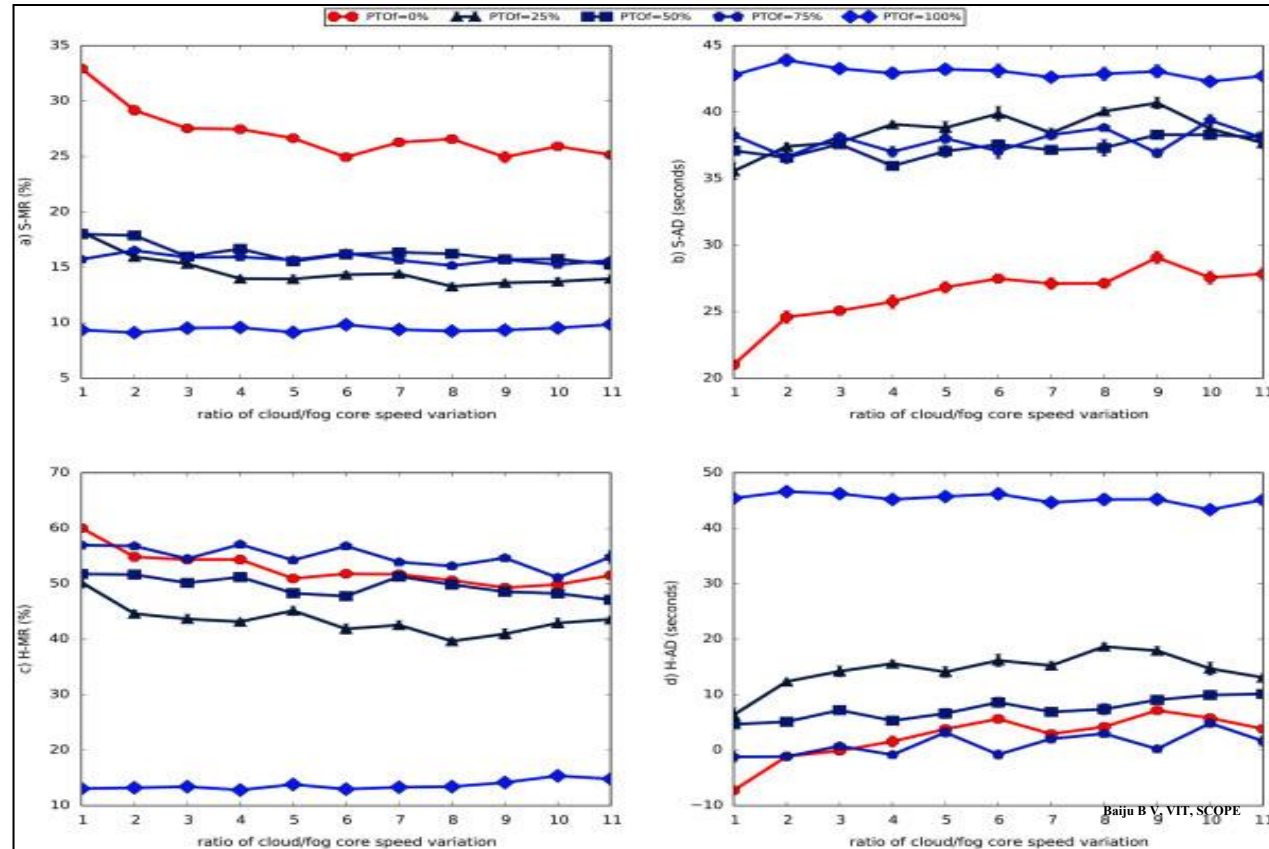
Integrated C2F2T Literature by Metrics

- Metrics observed in articles

Metric	Varlations
Energy consumption	Devices power consumption Energy efficiency Percentage energy saved
Performance	Latency Effective arrival rate Average system response System effectiveness Cache performance
Resource consumption	Usage of devices Average buffer occupancy
Costs	Itemized costs Operational costs
QoS	Quality of pattern recognition in images Response time Reliability
Security	Opacity

1.1 Energy Consumption

- IoT devices are limited by power source, typically a battery, which comprises application performance.
- W. Li, I. Santos et al. (System modelling and performance evaluation of a three-tier cloud of things)** proposed a model to represent a C2F2T architecture.
- They evaluated the energy consumption in each of the **application layers** (IoT, fog, and cloud), identified the main sources of consumption of each, and verified the energy consumption in relation to the increase in the number of devices in the architecture.



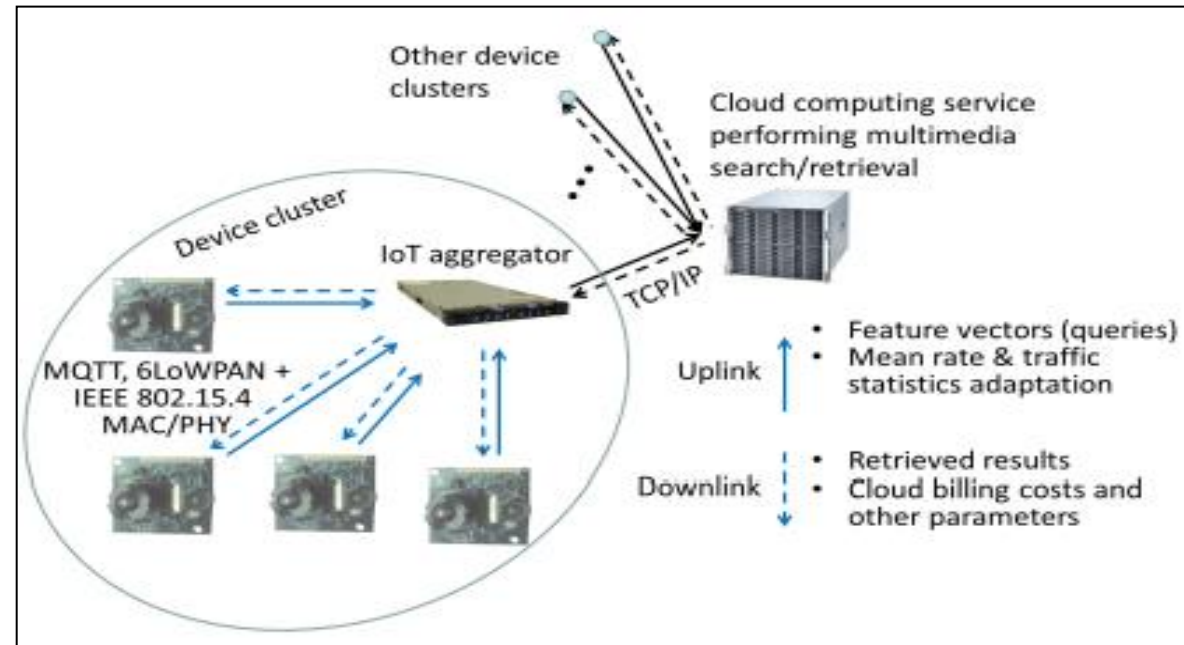
- **F. Renna** (Query processing for the Internet-of-Things: Coupling of device energy consumption and cloud infrastructure billing), presented energy consumption of devices in an IoT infrastructure in a different way.
- The model represents the device energy cost in three modes:
 - **Active**
 - **Idle**
 - **when the application switches from idle to active state.**

Smart Light:

Active Mode: The light is on, illuminating the room.

Idle Mode: The light is off but ready to turn on when needed.

Switching: The light turns on when someone enters the room.



- **L. Skorin-Kapov** (Energy efficient and quality-driven continuous sensor management for mobile IoT applications) address the energy consumption of idle and active devices sending data to a cloud environment, but in this work, the energy consumption for **transmitting**, **receiving**, **listening**, **sensing**, and **computing** are taken into account.

Traffic Monitoring:

Transmitting: Traffic sensors send data about vehicle flow to the cloud.

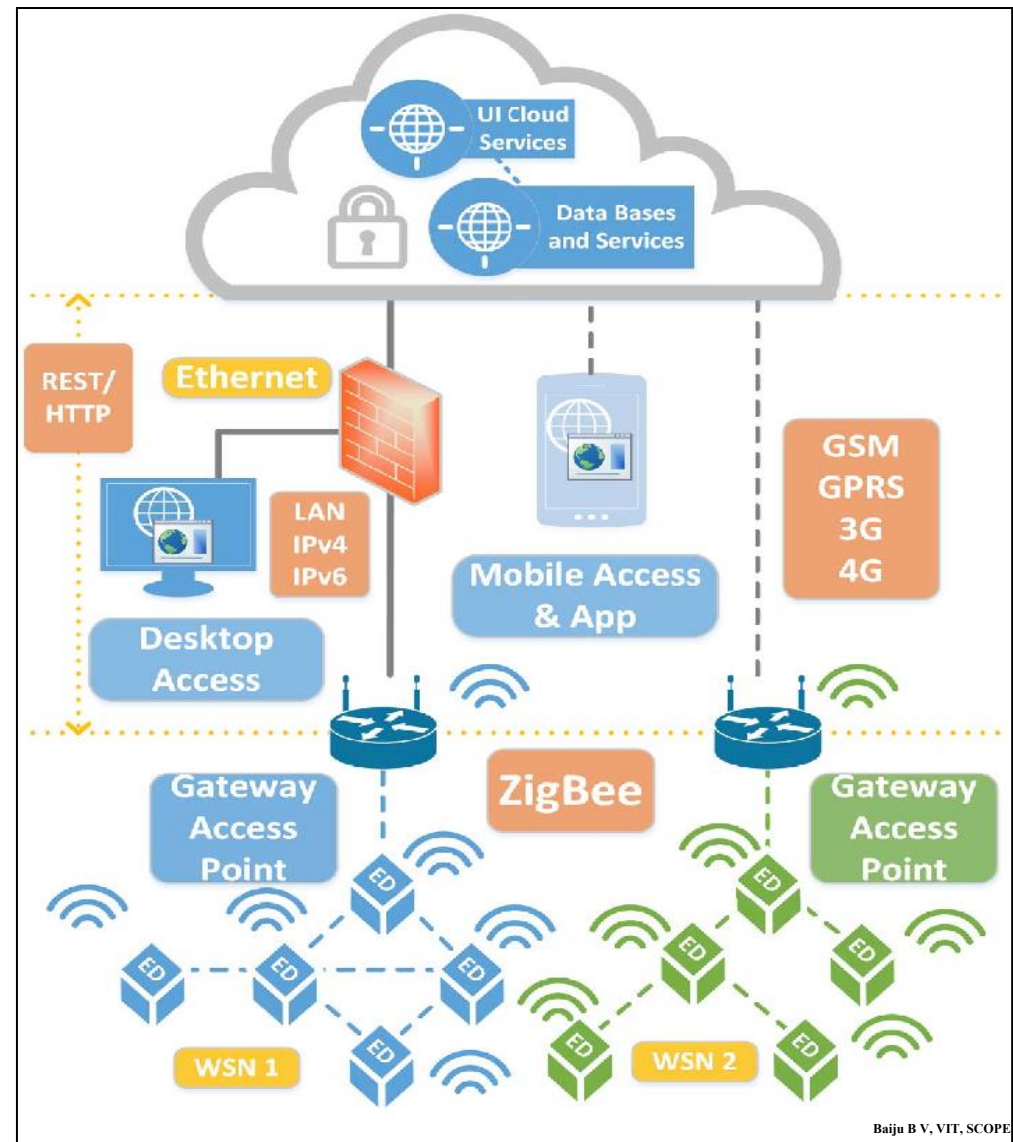
Receiving: Sensors receive updates or commands from the cloud to adjust their monitoring frequency.

Listening: Sensors stay alert for any emergency commands.

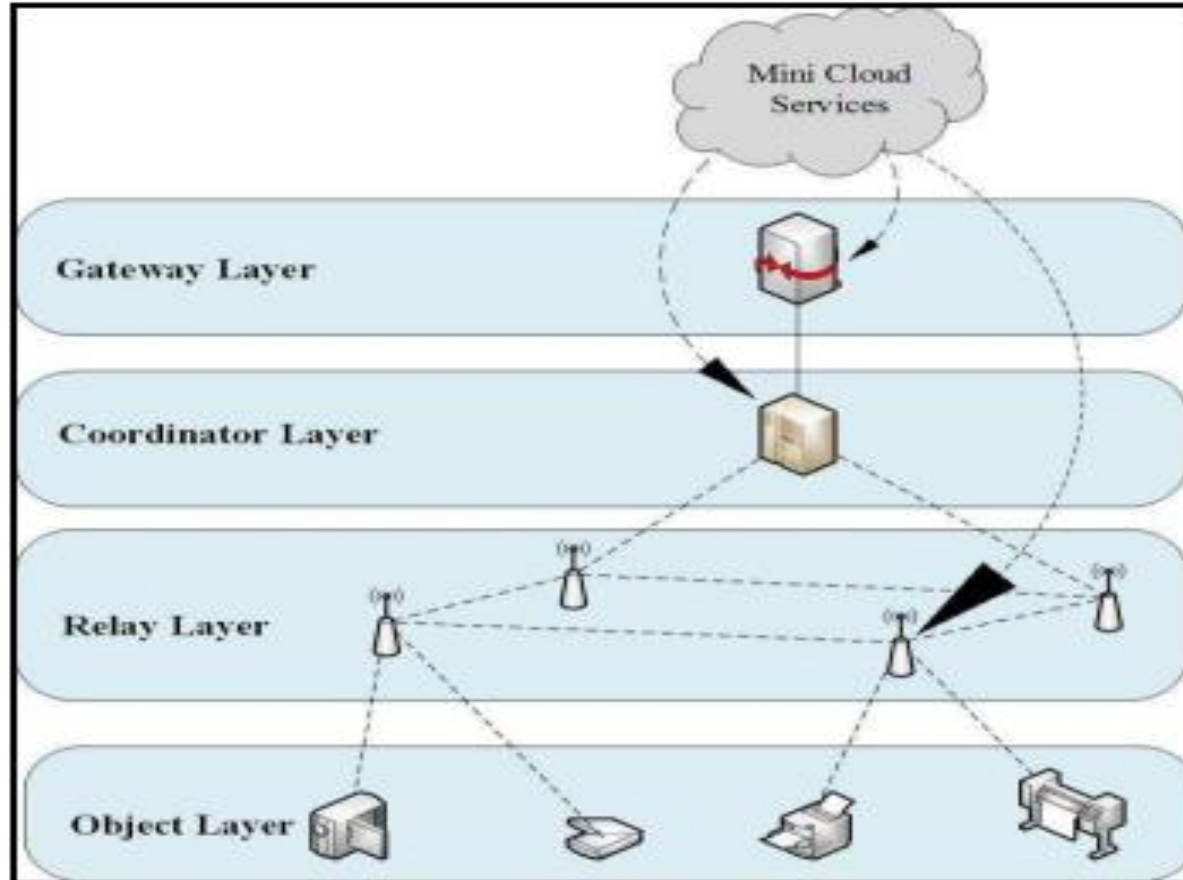
Sensing: Actively collecting data on traffic conditions.

Computing: Processing data locally to detect traffic anomalies before sending it to the cloud.

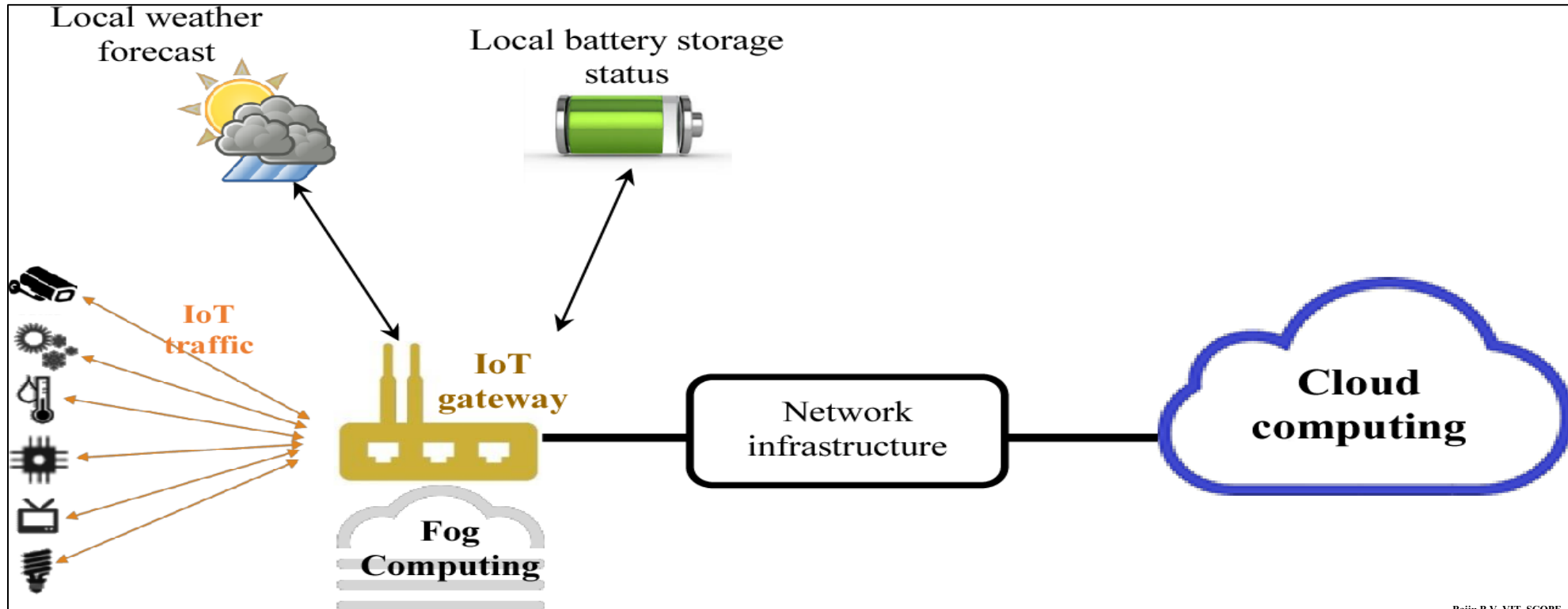
- **T. Gomes** (An IoT-based system for collision detection on guardrails) , considered the **operational mode of devices**, however, in this work the energy consumption depends on the **distance between the communicating nodes and the number of bits that will be transmitted**.
- In addition, the authors take into account the **medium** where the communication will occur – for instance, open space.



- **Zaineb T (Virtualization framework for energy efficient IoT networks)** addresses that the energy consumption is also impacted by processing-induced VMs located in the networking elements at the upper three layers.



- **Jalali et al. (Interconnecting fog computing and microgrids for greening IoT)** evaluate the energy consumption with respect to two perspectives:
 - the synchronization between IoT devices and the fog
 - the synchronization between IoT devices and the cloud.



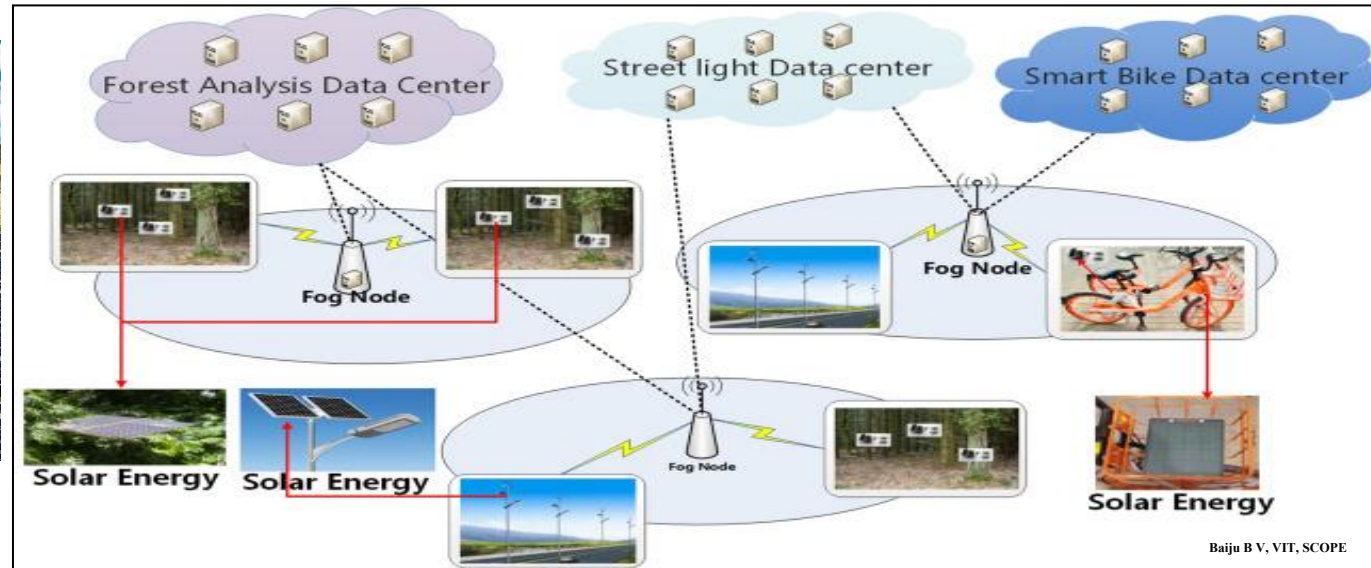
Synchronization between IoT devices and the fog

Consider a smart traffic management system where IoT sensors on traffic lights and vehicles communicate with nearby fog nodes. These fog nodes process data locally to make real-time decisions, such as adjusting traffic light timings to reduce congestion. The synchronization ensures that data from multiple sensors is processed simultaneously, enabling quick and efficient traffic management¹.

Synchronization between IoT devices and the cloud

Imagine a smart home system where various IoT devices like thermostats, security cameras, and smart locks send data to a cloud server. The cloud server processes this data to provide insights and control options via a mobile app. Synchronization ensures that all devices are updated with the latest settings and data, allowing users to control their home environment seamlessly from anywhere

- **Y. Li et. al. (Leveraging Renewable Energy in Edge Clouds for Data Stream Analysis in IoT assesses)** the energy consumption of **fog devices and cloud equipment**.
- The fog devices are more similar to the IoT devices with respect to **computational capacity**.
- It presents a renewable energy source where the fog devices are powered by photovoltaic panels.
- They evaluate the consumption of energy produced for photovoltaic panels, and other energy sources, such as batteries.



1.2 Performance

- **W. Li et al.** (System modelling and performance evaluation of a three-tier cloud of things) and **S. Sarkar et al.** (Theoretical modelling of fog computing: a green computing paradigm to support IoT applications) evaluated the system performance with respect to the **latency** of the application.
- This latency can be divided into
 - (i) **Processing latencies**
 - (ii) **Transmission latencies.**
- The processing latency is the time taken by an application to process all tasks.

This is the time your email application takes to prepare the email, attach the file, and get it ready to send. For instance, if your computer takes 2 seconds to attach a file and another 1 second to encrypt the email, the total processing latency is 3 seconds.

- The transmission latency is the communication delay to send a unit of data to the destination.

This is the time it takes for the email to travel from your computer to the recipient's email server. If your internet connection takes 5 seconds to send the email to the server, then the transmission latency is 5 seconds.

- **T. Dinh** (A location-based interactive model for Internet of Things and cloud (IoT-cloud)), evaluated the **packet delivery latency**
- This is calculated by
 - the number of hops to the destination
 - the sleep interval of a node
 - the time to transmit a data packet

Consider a small IoT network where a data packet needs to be sent from a sensor node to the cloud server. The packet must pass through 3 intermediate nodes (3 hops). Each intermediate node has a sleep interval of 20 ms, and the time to transmit a data packet per hop is 10 ms.

Sleep interval for each hop = 20 ms

Transmission time for each hop = 10 ms

Total hops = 3

Total Latency=Number of Hops × (Sleep Interval + Transmission Time)

Total Latency = $3 \times (20\text{ms} + 10\text{ms})$

Total Latency = $3 \times 30\text{ ms}$

Total Latency = 90 ms

- **E. Sturzinger** (Application-aware resource provisioning in a heterogeneous Internet of Things) presented the delay of processing applications requests which is defined analytically as a combination of **computational complexity of devices** and the **average flow size of application**.
- **Lyu et al.** (High-performance scheduling model for multisensor gateway of cloud sensor system-based smart-living) evaluate the performance of a **scheduling mechanism**.
 - For this, they assess the **scalability of the scheduling** and **caching solver** and add several sensors connected to cloud servers.
 - In addition, the **average waiting time** and **throughput** applications are analyzed.
- **G. Merlino et al.** (Mobile crowdsensing as a service: a platform for applications on top of sensing clouds) proposed a mobile crowdsensing scenario is used to evaluate the **performance** and the **effectiveness** of the system using a **Petri Net model**.

- **K. E. Desikan** (A Novel Distributed Latency-Aware Data Processing in Fog Computing-Enabled IoT Networks), examined the performance of C2F2T applications using a variety of means.
- The response time of system is defined as the time elapsed from the data was generated until it is processed by a gateway.
- Efficiency processing is formulated mathematically and takes account of
 - *the time to process the application data*
 - *the occupancy buffer*
 - *the response time of all gateways*

$T_p = 30\text{ms}$ (Time to process the application data).

$T_b = 20\text{ ms}$ (Buffer occupancy time).

$T_g = 50\text{ ms}$ (Total response time of all gateways).

$$\text{Total system response time } (T_{\text{total}}) = T_p + T_b + T_g$$

$$T_{\text{total}} = 30\text{ms} + 20\text{ms} + 50\text{ms}$$

$$T_{\text{total}} = \mathbf{100\text{ ms}}$$

1.3 Resource Consumption

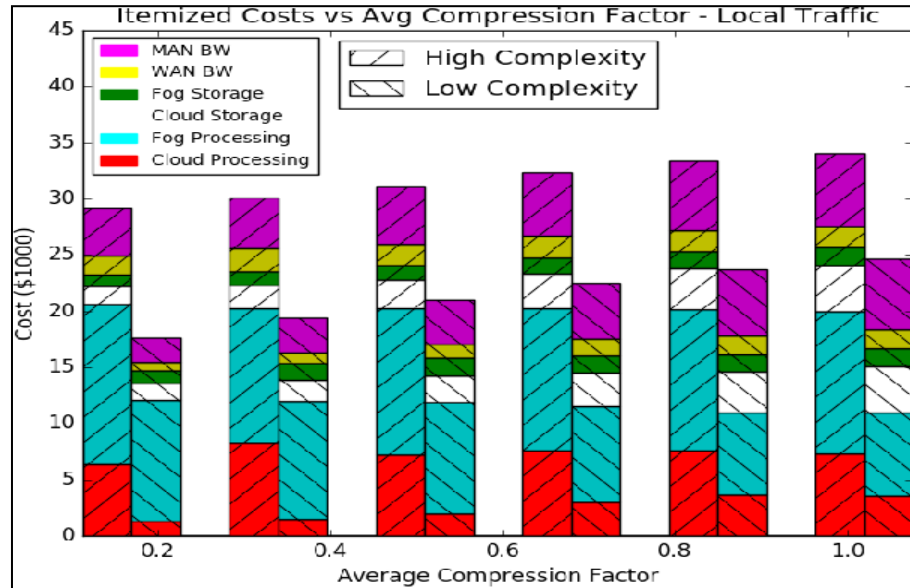
- Due of the limited capabilities of IoT devices, careful attention is required when allocating tasks on these devices.
- Fog and cloud computing offer additional capabilities for IoT devices, it is necessary to examine where such workloads should be processed.
- **V. B. Souza et al. (Towards Distributed Service Allocation in Fog-to-Cloud (F2C) Scenarios)** aim to obtain the **optimal allocation of services** on the available resources, taking into account resources of devices available in C2F2T infrastructures.
 - They present the advantages of using fog computing and evaluate the number of allocated resources and the usage of resource devices.
- **Y. Liu et al. (Incentive mechanism for computation offloading using edge computing: a Stackelberg game approach)** assessed the level consumption in C2T scenario.
- They evaluated the consumption about two aspects:
 - **Cloud infrastructure consumption**
 - **IoT devices consumption**

1.4 Cost

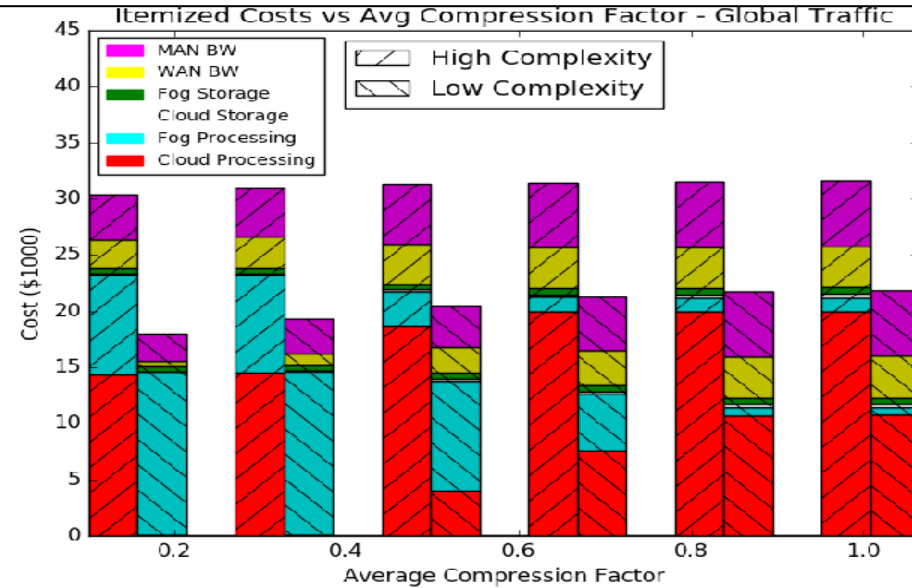
- Applications that require a lot of cloud computing resources can dramatically increase the cost to the service provider and ultimately the consumer
- **Renna et al. (Query processing for the Internet-of-Things: Coupling of device energy consumption and cloud infrastructure billing)** examine the **billing costs** when computational resources are reserved to process queries uploaded by IoT applications.
- Billing costs are directly proportional to the expected query volume generated by IoT devices.

In a smart agriculture system, IoT sensors monitor soil moisture, temperature, and humidity, continuously sending data to the cloud. If the number of sensors increases or the frequency of queries rises, the cloud infrastructure usage also increases. As a result, billing costs grow since they are directly tied to the volume of queries processed. This highlights the importance of optimizing both query frequency and resource allocation to control costs while maintaining efficiency.

- **Sturzinger et al.** (Application-aware resource provisioning in a heterogeneous Internet of Things) attempt to **minimize the operational cost** of provisioning IoT traffic to the cloud, while satisfying all application constraints.
- The **total cost** is composed of the sum of the cost of all devices including as **processing, storage, upstream** and **downstream cost, local** and **global traffic**, and the **link capacity cost**.



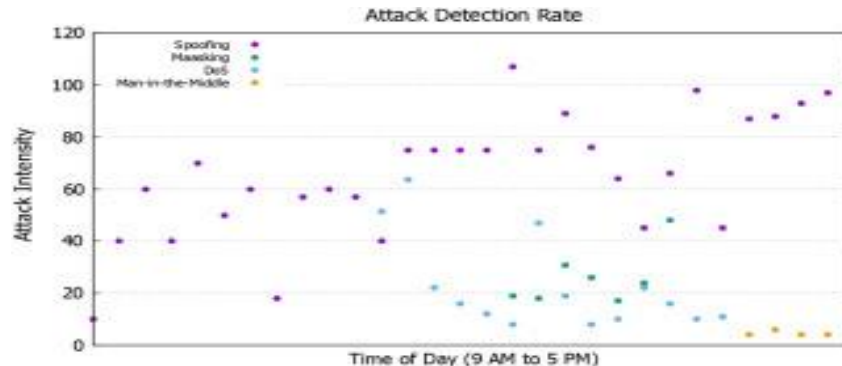
(a) Local traffic



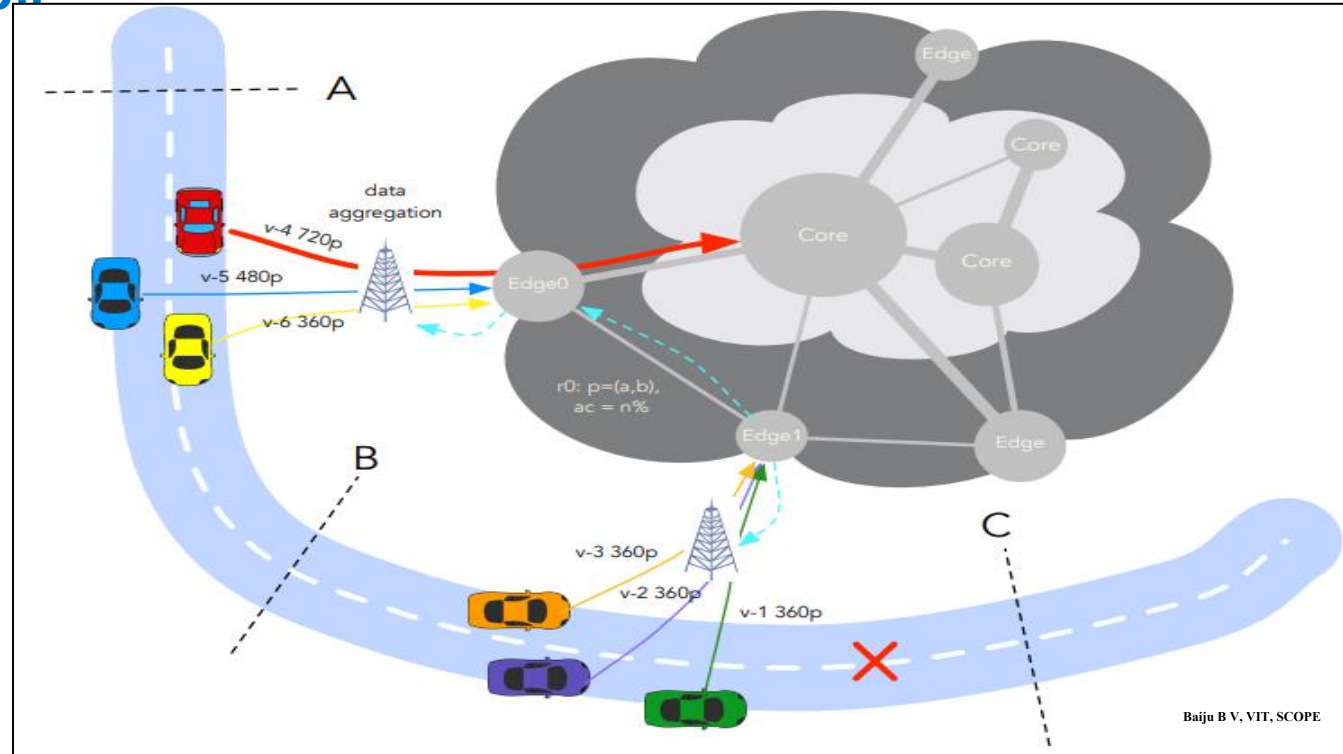
(b) Global traffic

1.5 Quality of Service

- **Z. Ming** (QoS-aware computational method for IoT composite service), evaluated **composite services** in an IoT environment.
- **R. K. Behera** (Reliability modelling of service-oriented Internet of Things) presented **reliability** of the IoT system depends on a variety of factors, including the
 - **availability of the service**
 - **availability of the input files to service**
 - **reliability of each sub-system** that composes the overall IoT system.
- **R. K. Lomotey** (Wearable IoT data stream traceability in a distributed health information system) defined reliability as the ability of the **traceability model** to detect and prevent attacks on applications.



- **Y. Li et al. (Leveraging Renewable Energy in Edge Clouds for Data Stream Analysis in IoT)** presented the **use case for analyses** of video streams in the cloud generated by vehicles on a road.
- Authors represented QoS as the **detection accuracy** of an object in video images.
- They vary the resolution of video in order to decrease the **CPU**, **memory** and **bandwidth consumption**
- In addition, considering the scenario where there are $2n+1$ cars, the accuracy is defined as the probability of a result (object detected in image) appears exceeds $n + 1$ times among $2n + 1$ results.



1.6 Security

- **R. K. Behera et al. (Reliability modelling of service-oriented Internet of Things)** evaluated the concept of **opacity in data flow** in the C2T context.
- Opacity is a uniform approach to describe security properties expressed as predicates.
- The authors used analytical models to verify the opacity in the medical research application scenario.